

Summary of public comments

Considerations for the Regulation of Generative AI-Enabled Medical Devices
Docket FDA-2026-N-7874

124 tagged comments, 2796 tags, 1711 skips, 5 duplicates not tagged · 30 sections · generated 09 October 2026

Each section says what commenters wrote about one part of FDA's paper. The line under a section's heading counts the comments and tags behind it, and the positions those tags take.

A number in brackets, such as [\[3\]](#), points to that section's sources, listed under its text. A source names a comment and a paragraph: 0126 ¶p091 is paragraph p091 of comment FDA-2026-N-7874-0126, so every claim can be looked up in the commenter's own words.

GENERAL: Overall views, background and definitions

81 comments, 342 tags: 58 support, 166 conditional-minor, 69 conditional-major, 6 oppose, 43 neutral

GEN: Overall approach and general views

75 comments, 246 tags: 53 support, 120 conditional-minor, 53 conditional-major, 5 oppose, 15 neutral

Of the 246 points, 53 support FDA's approach as it stands, 120 ask for small changes, 53 for larger ones, 5 oppose it and 15 take no position. Support centres on FDA's early, open engagement before any draft guidance [1], and on the competency-based, two-axis, lifecycle direction itself, which several commenters support or call sound [2]. Several endorse a risk-proportionate, least-burdensome lifecycle approach [3], and one says GenAI changes the scale of verification but not the principles [4].

The five opposing points come from individuals [5]. One cancer patient calls any healthcare use of generative AI too risky now [6]; another would bar such devices until bias and liability questions are settled [7]; a third warns of "grievous harm" from unconstrained AI [8]; a fourth opts out altogether [9]. The fifth would instead let market forces, not regulation, rein AI in [10].

The larger changes asked for point one way: beyond a point-in-time competency test to the deployed system across its life [11]. Proposals include an auditable lifecycle assurance case binding configuration, authority boundaries and change decisions [12]; behaviour that stays attributable and reconstructable as models and tools change [13]; a minimum runtime governance evidence package for higher-risk devices [14]; a mandatory execution-time gate asking whether the system is authorized to act now [15]; and a clinician-style licence with staged, earned autonomy [16]. One commenter says the paper neglects how devices are operated, secured, maintained, repaired and retired [17]; another warns that flexibility outruns the infrastructure, such as trusted benchmarks, needed to make it safe [18]; a third rates the paper strong conceptually but weak in evidentiary maturity, since competency need not predict clinical benefit [19].

Others ask FDA to build on machinery that already exists rather than a parallel track: ISO 14971 files, design and change controls [20], terminology FDA already uses elsewhere, such as established conditions [21], and existing risk-based regulation evolved into a lifecycle model [22]. A related call is for technology neutrality, regulating clinical function, risk and autonomy rather than the "generative" label [23]; one commenter wants autonomy treated as a cross-cutting topic of its own [24].

Human accountability is another thread [25]: one commenter says accuracy belongs to models and accountability to people [26]; another urges preserving physician oversight of consequential decisions [27]; a third says "human involvement" offered as a mitigation needs the same evidentiary rigour as the AI itself [28]. Cautions include that least-burdensome goals must not displace FDA's safety mandate [29], that economic incentives may push autonomy past the evidence [30], and that no GenAI device has yet been authorized, so the framework is an untested scaffold [31].

Process requests include moving to draft guidance [32], a published pathway decision matrix [33], naming which proposals need new legal authority [34], international harmonization including through IMDRF [35], pilots and workshops [36], explicit decision criteria in future guidance [37], and relief for small or early-stage companies [38]. Among neutral points, one commenter notes the paper's consistency with existing FDA, IMDRF, WHO and NIST guidance [39], and another predicts, from gene therapy's history, that AI regulation will be reactive [40].

Sources. [1] 0003 ¶p009; 0013 ¶p185; 0028 ¶p001; 0080 ¶p004; 0100 ¶p005; 0126 ¶p297 [2] 0006 ¶p002; 0051 ¶p009; 0058 ¶p002; 0087 ¶p048; 0093 ¶p004; 0099 ¶p026; 0124 ¶p020 [3] 0030 ¶p001; 0035 ¶p006; 0048 ¶p001 [4] 0074 ¶p007 [5] 0007 ¶p001; 0011 ¶p004; 0034 ¶p002; 0041 ¶p001; 0031 ¶p017 [6] 0007 ¶p001 [7] 0011 ¶p004 [8] 0034 ¶p002 [9] 0041 ¶p001 [10] 0031 ¶p017 [11] 0035 ¶p005; 0049 ¶p006; 0115 ¶p004; 0105 ¶p010; 0045 ¶p006 [12] 0035 ¶p005 [13] 0049 ¶p006 [14] 0115 ¶p004, ¶p005, ¶p006 [15] 0105 ¶p010, ¶p011 [16] 0045 ¶p006 [17] 0032 ¶p001 [18] 0013 ¶p186 [19] 0020 ¶p004, ¶p005 [20] 0013 ¶p175, ¶p176; 0093 ¶p006 [21] 0067 ¶p035 [22] 0087 ¶p049, ¶p050, ¶p051 [23] 0087 ¶p009; 0126 ¶p240, ¶p241; 0045 ¶p114 [24] 0126 ¶p243 [25] 0046 ¶p014; 0097 ¶p013; 0098 ¶p046 [26] 0046 ¶p014 [27] 0097 ¶p013, ¶p014 [28] 0098 ¶p046 [29] 0107 ¶p005 [30] 0097 ¶p009, ¶p010 [31] 0106 ¶p019 [32] 0012 ¶p012 [33] 0003 ¶p089 [34] 0013 ¶p181, ¶p182; 0116 ¶p023 [35] 0003 ¶p014; 0024 ¶p010; 0038 ¶p005; 0051 ¶p009 [36] 0035 ¶p130 [37] 0099 ¶p027 [38] 0003 ¶p083, ¶p084, ¶p085; 0094 ¶p020, ¶p021, ¶p022 [39] 0054 ¶p032, ¶p033, ¶p034, ¶p035, ¶p036 [40] 0109 ¶p009

BKG: GenAI characteristics, risks and benefits

19 comments, 30 tags: 5 support, 1 oppose, 24 neutral

Most of this section is descriptive: 24 of the 30 points set out GenAI's traits, benefits or risks without taking a position, 5 endorse the paper's account of them and 1 opposes the devices outright.

Those endorsing say the paper correctly names open-ended inputs, variable outputs, frequent change and reliance on third-party models [1]; two of them add that this rightly makes static or point-in-time validation insufficient on its own [2]; one adds that studies of planted false details and low diagnostic accuracy make its account of confabulation and drift, if anything, understated [3]. The opposing point calls the devices "hugely problematic" because narrow training data can build in bias against races, ages or other groups [4].

Hoped-for benefits include wider access to specialist knowledge and current evidence, less clinician burnout, more consistent care and earlier detection of risk [5]. One rural hospital worker expects AI reading of head CTs to speed stroke transfers if other facilities and insurers accept it [6], and one commenter predicts AI will err far less often than humans [7].

Risks described include a chatbot's bromide-for-salt suggestion that put a man in hospital [8], failure through omission, harmful agreement, drift and escalation that fails in practice [9], sycophancy traced to training for agreeableness [10], answering the question asked rather than the need behind it [11], characteristic errors in model-written nursing care plans [12], and output variability that persists in generative serving whatever the determinism settings, so reproducibility must be engineered and shown [13].

Sources. [1] 0020 ¶p011; 0029 ¶p003, ¶p004; 0038 ¶p001, ¶p004 [2] 0020 ¶p011; 0029 ¶p004 [3] 0106 ¶p011 [4] 0011 ¶p001 [5] 0012 ¶p002; 0017 ¶p002; 0021 ¶p006; 0025 ¶p012, ¶p450, ¶p454; 0086 ¶p002 [6] 0036 ¶p001 [7] 0031 ¶p015 [8] 0023 ¶p002 [9] 0021 ¶p005 [10] 0061 ¶p031, ¶p033 [11] 0025 ¶p013, ¶p014 [12] 0121 ¶p046 [13] 0126 ¶p206, ¶p246

DEF: Definitions, scope and device status

18 comments, 66 tags: 46 conditional-minor, 16 conditional-major, 4 neutral

Of the 66 points, 46 ask for small changes, 16 for larger ones and 4 take no position; none supports or opposes.

A recurring request is that FDA say where a GenAI device function begins and ends [1]. One calls location an unreliable boundary and would characterize the function by properties such as purpose, recipient, directiveness, review opportunity and authority to act [2], illustrated by services that turn into device functions as they infer or act [3]. Another wants a consistent line between AI supporting a clinician's plan and AI exercising independent judgment [4]; another, examples and boundary controls where one foundation model serves regulated and non-regulated functions [5]. One commenter warns that administrative-seeming agents can shape care, asks who regulates AI that is not a device, and

proposes an influence-based principle with federal and state partners [6]; another would adopt a "developer-to-user" view like farm-to-fork food safety [7].

Several commenters want low-risk informational or supervised functions kept outside active regulation [8]. One asks that informational, non-directive functions be non-devices or under enforcement discretion, with a bright line against the wellness policy and one standard whoever deploys the model [9]. Another would default to enforcement discretion and the existing decision-support exclusion for clinician-facing informational functions [10]; another would tie the lower-risk clinician-facing class to the independent-review criterion [11]. One joint comment requests narrow enforcement discretion for image-analysis functions that only draft private, unverified text for the interpreting radiologist, with reliance before specialist verification as the boundary [12].

Others push back: a professional society calls the non-device decision-support four-point test arbitrary now that chatbots analyze images and signals, and would study exempt tools [13]; one commenter suggests AI therapists claiming to treat psychiatric conditions be regulated like devices [14].

On definitions and classification, one commenter asks for consensus terms across agencies, since NIST and FDA definitions differ [15]; another, whether insight-deriving devices and causal AI count as generative [16]; another, how the approach applies to combination products another center leads [17]; another, attention to functions whose output is a patient-specific implant [18]. One would keep classification by intended use, with substantial equivalence turning on characteristics so that open-ended output is not equivalent to a bounded predicate [19].

Sources. [1] 0035 ¶p015; 0072 ¶p006; 0018 ¶p006; 0120 ¶p099 [2] 0035 ¶p015, ¶p016 [3] 0035 ¶p028, ¶p029, ¶p030, ¶p031, ¶p032, ¶p033 [4] 0072 ¶p006 [5] 0018 ¶p003, ¶p006, ¶p007 [6] 0025 ¶p020, ¶p021, ¶p295, ¶p443, ¶p482 [7] 0038 ¶p010, ¶p011 [8] 0080 ¶p005; 0094 ¶p023; 0120 ¶p011; 0046 ¶p016 [9] 0080 ¶p005, ¶p009, ¶p010, ¶p011, ¶p012, ¶p013, ¶p014 [10] 0094 ¶p023 [11] 0046 ¶p003, ¶p016 [12] 0120 ¶p011, ¶p012, ¶p061, ¶p063, ¶p065, ¶p099 [13] 0116 ¶p020, ¶p007 [14] 0023 ¶p003 [15] 0038 ¶p008, ¶p009 [16] 0069 ¶p001 [17] 0067 ¶p012, ¶p013 [18] 0053 ¶p005 [19] 0126 ¶p249, ¶p250, ¶p251

RISK: Assessing a device's risk

72 comments, 516 tags: 31 support, 290 conditional-minor, 148 conditional-major, 7 oppose, 40 neutral

Q1: Two-axis risk framework and additional dimensions

64 comments, 251 tags: 15 support, 88 conditional-minor, 119 conditional-major, 2 oppose, 27 neutral

Of the 251 points, 15 support the two-axis framework as it stands, 88 ask for small changes, 119 for larger ones, 2 oppose it and 27 take no position. Supporters call the two axes the right organizing principle, clinically intuitive, or a sound heuristic that should be kept rather than turned into an unmanageable multidimensional matrix [1]. The two opposing points come from one commenter, who says the framework does not capture GenAI risk and is "not even needed" [2].

Commenters take up each of the four dimensions FDA itself floated. Reversibility is asked for as a first-order factor, a gating step before a function's final risk position, or a disqualifier from reduced premarket evidence [3]. Time pressure and downstream safeguards are wanted as explicit modifiers or first-class dimensions, one commenter noting that on a vessel days from port the device is the last safeguard [4]. Traceability to source material is wanted as a verifiable mitigating factor, since inspectable outputs let errors be caught [5]. Commenters differ on form: several would keep two axes and record the extra factors as documented modifiers [6], while others propose four axes, five axes, a three-layer model adding model and lifecycle risks, or four dimensions organized around whether a function presents or generates a clinical claim [7].

A second theme is that the axes capture severity but not probability. Several commenters want the probability that an incorrect output escapes detection, or error detectability, added, and the grid reconciled with ISO 14971 rather than run as a parallel taxonomy [8].

A third theme is human oversight. Several commenters say a clinician nominally "in the loop" should lower risk only when the reviewer has the competence, information, authority and time to intervene, with that effectiveness demonstrated under realistic conditions rather than asserted [9]. Others would make a named accountable reviewer an explicit risk-lowering variable [10], or base activity-axis placement on what the generative component is architecturally permitted to write and initiate [11].

Commenters also argue that the consequence axis, limited to "incorrect output", misses harms from correct ones [12]: a navigation agent that retrieves an insurance card for a man having a heart attack [13], omission and harmful affirmation [14], a device talked out of its concern by the user [15], and relational features such as memory and avatars [16]. Further candidate dimensions include data exposure or sensitivity [17], the counterfactual care available and duration of reliance [18], deployment context in under-resourced settings [19], third-party model dependency and change dynamics [20], connectivity and fleet-wide common-mode failure [21], attribution of an action to an accountable person [22], architectural containment of agentic systems [23], and physical implant output [24]. One commenter would add execution-time authorization integrity as a pass, hold or escalate gate [25].

Neutral entries supply illustrations [26]: worked risk profiles under a four-dimension scheme [27], state laws under which the same device sits at different supervised positions [28], and a taxonomy of assistive, augmentative and autonomous software [29].

Sources. [1] 0003 ¶p016; 0020 ¶p012; 0049 ¶p023, ¶p024; 0076 ¶p024; 0087 ¶p011 [2] 0019 ¶p003, ¶p020 [3] 0003 ¶p019; 0099 ¶p009; 0118 ¶p007; 0131 ¶p016, ¶p024 [4] 0003 ¶p020; 0028 ¶p021; 0082 ¶p016; 0119 ¶p028 [5] 0009 ¶p011; 0087 ¶p014; 0094 ¶p033; 0100 ¶p022 [6] 0051 ¶p010; 0054 ¶p054; 0049 ¶p027; 0055 ¶p021 [7] 0015 ¶p003; 0045 ¶p017; 0038 ¶p014; 0126 ¶p060,

¶p066 [8] 0013 ¶p009, ¶p025, ¶p026; 0020 ¶p021; 0055 ¶p014; 0052 ¶p007 [9] 0017 ¶p047; 0021 ¶p016; 0025 ¶p400; 0058 ¶p031; 0087 ¶p013; 0098 ¶p028; 0099 ¶p013; 0120 ¶p133 [10] 0046 ¶p015; 0129 ¶p007 [11] 0127 ¶p029; 0103 ¶p012 [12] 0025 ¶p036; 0021 ¶p009; 0111 ¶p011 [13] 0025 ¶p029, ¶p035, ¶p036 [14] 0021 ¶p009 [15] 0111 ¶p011 [16] 0021 ¶p022 [17] 0076 ¶p032; 0101 ¶p004 [18] 0006 ¶p019 [19] 0094 ¶p034 [20] 0003 ¶p018; 0093 ¶p010 [21] 0032 ¶p043 [22] 0042 ¶p005 [23] 0037 ¶p003 [24] 0053 ¶p007 [25] 0105 ¶p021 [26] 0126 ¶p339; 0119 ¶p026; 0121 ¶p066 [27] 0126 ¶p339 [28] 0119 ¶p026 [29] 0121 ¶p066

Q2: Directiveness spectrum for informational functions

31 comments, 51 tags: 6 support, 29 conditional-minor, 12 conditional-major, 2 oppose, 2 neutral

Of the 51 points, 6 support FDA's positions as they stand, 29 ask for small changes, 12 for larger ones, 2 oppose and 2 take no position. FDA's two positions, that directiveness turns on substance and context rather than words such as "recommend", and that a "talk to your doctor" disclaimer does not make an output less directive, are endorsed outright by several commenters [1] and accepted by others who then ask for more [2].

Two kinds of small change recur. One names the output characteristics that should modify risk: specificity, personalization, imperative force or urgency, displayed confidence, suppressed or omitted alternatives, repetition, who receives the output and whether it enters records or ordering systems [3]. The other asks for predictability: worked examples along the continuum, a rubric with anchored levels, a decision tree or checklist, a rebuttable presumption that outputs with numeric parameters are action-directing, or a duty to document where each output type sits and test for drift [4].

Larger changes would move away from grading wording at all. One commenter would define the axis by a function's effect on the user's choice set, from presenting to executing, with a rung for omission [5]; another would use a single division between general and patient-specific output [6]; another calls directiveness an emergent property sponsors cannot attest to and would have them specify a ceiling with a measured escape rate [7]; another would measure it by how often clinicians override outputs [8]. One opposing point says directiveness varied with the user's displeasure in the record, so it cannot be a modifier [9].

Two commenters add that for emergencies a directive output is the safe one, so assessment should ask whether directiveness fits the situation [10]; one of them asks for a protocol-delivery sub-category judged by fidelity to a published protocol [11].

Sources. [1] 0020 ¶p013; 0038 ¶p017; 0076 ¶p024; 0126 ¶p050; 0127 ¶p020 [2] 0006 ¶p020; 0028 ¶p022; 0093 ¶p011; 0100 ¶p024; 0118 ¶p012 [3] 0045 ¶p025; 0048 ¶p003; 0049 ¶p030; 0052 ¶p033; 0120 ¶p136 [4] 0003 ¶p022; 0028 ¶p022; 0051 ¶p011; 0100 ¶p025; 0118 ¶p012 [5] 0131 ¶p035, ¶p040 [6] 0055 ¶p022, ¶p028 [7] 0013 ¶p030, ¶p031 [8] 0113 ¶p012 [9] 0065 ¶p031 [10] 0082 ¶p018; 0100 ¶p027 [11] 0082 ¶p022

Q3: Patient-facing vs clinician-facing functions

30 comments, 50 tags: 5 support, 37 conditional-minor, 7 conditional-major, 1 oppose

Of the 50 points, 5 support the paper's treatment as it stands, 37 ask for small changes, 7 for larger ones and 1 opposes. Supporters agree that patient-facing use can carry more risk when patients cannot assess outputs, and that the paper is right to warn against paternalism and against underestimating patients [1]. The opposing point says people should keep relying on human mental health professionals [2].

The recurring small change is that patient-facing status should not raise risk automatically. Several commenters would judge instead by the user's realistic path to verification, the enforceability of safeguards in the setting, or what the output could cause the patient to do [3]. Two would allow a

conditional upward shift, offset by demonstrated mitigations or limited to triage, dose and escalation outputs [4]. Two others want more weight on patient-facing risk [5]: one calls it unequivocally higher and rejects empowerment as a goal [6]; the other notes that patients misdescribe symptoms and use tools instead of care, so an absent clinician backstop should itself raise risk [7].

Safeguards commenters would credit include calibrated uncertainty communication, escalation triggers that cannot be suppressed by conversational drift, deferral to a human rather than ending the interaction, traceable sources, comprehension testing across health-literacy levels, limited scope, and disclosure that the user faces a machine [8]. One commenter building for bystander emergencies adds deterministic protocol selection, refusal that redirects to the nearest safe action, and offline operation [9].

Larger changes question the patient-versus-clinician split itself: a dedicated user dimension, since the user changes error detection rather than harm severity [10]; a third tier for trained, supervised non-clinical health workers [11]; and device-specific competency affirmation replacing category labels, since credentials do not confer the ability to evaluate outputs [12]. One commenter adds that licensure does not make clinicians reliable safeguards, as no practitioner can stay current with all evidence [13].

Sources. [1] 0003 ¶p023; 0021 ¶p024; 0055 ¶p032, ¶p033; 0126 ¶p098 [2] 0019 ¶p008 [3] 0006 ¶p021; 0013 ¶p034; 0030 ¶p005; 0045 ¶p027; 0049 ¶p032; 0052 ¶p048; 0054 ¶p057; 0100 ¶p028; 0115 ¶p015; 0120 ¶p141 [4] 0051 ¶p012; 0093 ¶p012 [5] 0043 ¶p009; 0118 ¶p014 [6] 0043 ¶p009 [7] 0118 ¶p014, ¶p015, ¶p016 [8] 0028 ¶p022; 0038 ¶p025; 0049 ¶p033; 0055 ¶p034; 0076 ¶p065; 0100 ¶p030, ¶p031, ¶p032, ¶p033 [9] 0082 ¶p023, ¶p024, ¶p025, ¶p026 [10] 0126 ¶p097 [11] 0009 ¶p014 [12] 0037 ¶p006, ¶p008 [13] 0025 ¶p067

Q4: Generalist vs specialist clinician users

19 comments, 25 tags: 21 conditional-minor, 2 conditional-major, 1 oppose, 1 neutral

Of the 25 points, 21 ask for small changes, 2 for larger ones, 1 opposes and 1 takes no position; none supports it. The shared small change is that professional title alone should not set risk. Several would ask instead whether the intended user can independently evaluate the output in the device's domain [1]. One illustrates with radiation oncology, where plausible output can hide unsafe fractionation or a missed prior course [2].

Several would make the intended user and expected training part of the intended use, validated in that population [3]. Testing proposals include simulated users matching the least specialized intended user [4], generalist-context benchmark data [5], and specialist confirmation where safe interpretation needs specialist knowledge [6]. Safeguards named include verified roles, role-based access, specialist confirmation, automatic deferral, referral prompting tested as a competency, output modifiers flagging specialist content, and workflow controls rather than labeling alone [7]. One adds that a disclaimer to consult a specialist is not an adequate safeguard [8]; another that specialist status lowers risk only with source access, finalization control and training [9]. Two stress the access benefit of moving knowledge to generalists, with escalation when needed [10].

The larger changes are to assess risk against foreseeable off-label users, crediting technical enforcement but not labeling [11], and to replace the credential distinction with device-specific competency affirmation [12]. The opposing point says no safeguard mitigates the risk; only keeping humans in place does [13].

Sources. [1] 0005 ¶p252, ¶p305; 0028 ¶p022; 0049 ¶p034; 0052 ¶p052; 0055 ¶p044; 0056 ¶p006; 0113 ¶p014; 0126 ¶p091 [2] 0056 ¶p007 [3] 0003 ¶p025; 0055 ¶p045, ¶p046; 0118 ¶p018; 0129 ¶p008 [4] 0051 ¶p013 [5] 0093 ¶p013 [6] 0056 ¶p009 [7] 0043 ¶p011; 0049 ¶p035; 0051 ¶p013; 0055 ¶p047, ¶p048, ¶p049, ¶p050; 0056 ¶p009 [8] 0056 ¶p008 [9] 0120 ¶p145, ¶p147 [10] 0025 ¶p321, ¶p325; 0051 ¶p013 [11] 0013 ¶p036 [12] 0037 ¶p009 [13] 0019 ¶p012

Q5: Multi-turn conversational risk

35 comments, 81 tags: 4 support, 65 conditional-minor, 4 conditional-major, 8 neutral

Of 81 points, 4 support, 65 ask for small changes, 4 for larger ones and 8 take no position. Supporters call trajectory-level assessment, with cumulative drift counted as a boundary failure, the right approach [1].

The small changes share a premise: the session or trajectory, not the turn, is the unit of intended use and evaluation; turn-level testing is insufficient alone [2]. Several would classify a device by the highest-risk output reachable along a foreseeable trajectory [3]. Several would define intended use as a declared envelope: permitted states and transitions, maximum directiveness, escalation conditions or the plausible endpoint [4]; one treats drift outside it as a reportable failure [5].

Testing proposals include adversarial simulation with personas such as symptom minimizers and escalation resisters [6]; long conversations, new sessions, stored memory, fictional framing, contradictory or delayed information, prompt injection and repeated runs [7]; detection of transitions, followed by warnings, tighter safeguards or revalidation [8]; and a drift check over cumulative state as a gate separate from per-turn tests [9]. One wants "multi-turn" to mean days and months, with results stratified by length [10]. On scoring, one would report timeliness and later recovery separately, so a late escalation is not credited [11]; another would score whether the device knew to ask another question [12]; a third counts failures to use context already collected [13]. Safeguards named include deterministic filters outside the model for boundaries such as age and user role [14], and enforceable scope restrictions, refusal, mandatory handoff and oversight checkpoints [15], with uncontrolled out-of-scope behavior treated as foreseeable [16].

One commenter reports a recorded session in which a document-drafting assistant produced an unsolicited directive, triggered by its reading of user displeasure rather than any request, while its safety markers improved and accuracy did not [17]. It asks that trajectory tests hold the request constant and vary affect, report unsolicited directives against turn index, include persistent state written across sessions, and demonstrate boundaries with the instruction layer absent [18]. Its larger change: a non-directive intended use that the device can exceed unasked does not describe it, so the statement must widen or the device be constrained [19]. Another larger change would make drift a logged gate refusal, characterizing intended use by what the gates permit [20]. The neutral entries describe that session and its limits [21], one commenter's own asymmetric risk estimate [22], and a clinician's experience with a tool that flags out-of-scope questions [23].

Sources. [1] 0020 ¶p014; 0076 ¶p003, ¶p025; 0127 ¶p021 [2] 0003 ¶p026; 0013 ¶p038, ¶p039; 0025 ¶p100; 0049 ¶p036; 0105 ¶p022; 0114 ¶p010; 0115 ¶p016; 0120 ¶p149 [3] 0003 ¶p027; 0052 ¶p053; 0065 ¶p034; 0126 ¶p107; 0131 ¶p041 [4] 0006 ¶p022; 0028 ¶p010; 0035 ¶p049; 0051 ¶p014; 0093 ¶p014; 0113 ¶p015; 0118 ¶p020; 0120 ¶p150 [5] 0051 ¶p014 [6] 0006 ¶p022; 0013 ¶p038; 0025 ¶p110; 0035 ¶p050; 0045 ¶p031; 0111 ¶p012; 0114 ¶p011 [7] 0021 ¶p014; 0045 ¶p031; 0055 ¶p054, ¶p055, ¶p056, ¶p057, ¶p058, ¶p059, ¶p060; 0105 ¶p024; 0115 ¶p016 [8] 0043 ¶p013; 0049 ¶p037; 0105 ¶p023 [9] 0098 ¶p012, ¶p032 [10] 0021 ¶p012, ¶p014 [11] 0091 ¶p039, ¶p040, ¶p041 [12] 0025 ¶p111 [13] 0088 ¶p029 [14] 0055 ¶p061, ¶p070 [15] 0052 ¶p054, ¶p056, ¶p057, ¶p059; 0120 ¶p151 [16] 0055 ¶p071 [17] 0065 ¶p010, ¶p011, ¶p017 [18] 0065 ¶p035, ¶p036, ¶p039, ¶p042 [19] 0065 ¶p013, ¶p041 [20] 0127 ¶p031 [21] 0065 ¶p015, ¶p044 [22] 0111 ¶p013 [23] 0019 ¶p014

Q6: Care escalation: under- vs over-escalation

36 comments, 58 tags: 1 support, 50 conditional-minor, 4 conditional-major, 1 oppose, 2 neutral

Of 58 points, 1 supports the paper as it stands, 50 ask for small changes, 4 for larger ones, 1 opposes and 2 take no position. The supporter calls recognizing both error directions more clinically mature than a single accuracy statistic [1]; the opponent says there are no acceptable trade-offs in any clinical

context [2].

Half the commenters make the same small change: never fold the two errors into one score, but report each separately [3]. Many would have the sponsor prespecify and justify its operating point or weighting for the deployment context [4], three adding that FDA should review that justification rather than set one threshold across settings [5]. Several want the asymmetry to favor safety, weighting serious under-escalation more heavily [6]; two would set very low tolerance for missing alarm or anaphylaxis presentations and bounded tolerance for over-referral [7]. Three push back [8]: over-escalation harms are diffuse and under-detected after market [9], can remove a veteran from care permanently [10], or, as errors of commission, can be the more dangerous [11].

Reporting proposals include stratifying by acuity or scenario severity [12], separate denominators with declared non-escalation cases in the corpus [13], comparison with clinician performance or documented rater disagreement [14], linked outcomes adjudicated by independent experts [15], and measures of timing, destination and handoff completion [16]. Four commenters question the binary framing [17]: escalation is graded by level, timing and destination [18]; a handoff can fail after it starts [19]; an intermediate pathway such as a telehealth nurse changes the balance [20]; for bystander protocols escalation is a step whose ordering is scored [21].

The larger changes are a third routing option to an accountable care team [22], a third error of escalating badly, with sponsors naming who responds and how fast [23], a pass-or-fail limit for severe under-escalation that common cases cannot offset [24], and treating the two errors as sensitivity and specificity for a defined condition rather than incommensurable [25].

Sources. [1] 0020 ¶p016 [2] 0019 ¶p016 [3] 0005 ¶p297; 0006 ¶p023; 0009 ¶p021; 0017 ¶p023; 0035 ¶p050, ¶p051; 0049 ¶p039; 0051 ¶p015; 0054 ¶p004; 0067 ¶p014; 0091 ¶p044; 0093 ¶p015; 0100 ¶p037; 0111 ¶p016; 0113 ¶p016; 0115 ¶p016; 0116 ¶p052; 0118 ¶p022; 0120 ¶p153 [4] 0006 ¶p023; 0009 ¶p021; 0013 ¶p041; 0026 ¶p003; 0028 ¶p011; 0045 ¶p033; 0049 ¶p039; 0051 ¶p015; 0067 ¶p015; 0076 ¶p061; 0084 ¶p003; 0091 ¶p047; 0093 ¶p015; 0100 ¶p036; 0111 ¶p016; 0120 ¶p155 [5] 0028 ¶p011; 0084 ¶p003; 0093 ¶p015 [6] 0003 ¶p028; 0006 ¶p023; 0026 ¶p002; 0054 ¶p048; 0076 ¶p061; 0100 ¶p036; 0118 ¶p022 [7] 0006 ¶p023; 0118 ¶p022 [8] 0013 ¶p043; 0043 ¶p015; 0111 ¶p015 [9] 0013 ¶p043 [10] 0111 ¶p015 [11] 0043 ¶p015 [12] 0054 ¶p065; 0100 ¶p037; 0118 ¶p022 [13] 0091 ¶p046 [14] 0006 ¶p023; 0100 ¶p038 [15] 0116 ¶p052 [16] 0055 ¶p078, ¶p079, ¶p080, ¶p081 [17] 0026 ¶p003; 0052 ¶p067; 0055 ¶p076; 0082 ¶p027 [18] 0052 ¶p067 [19] 0055 ¶p076 [20] 0026 ¶p003 [21] 0082 ¶p027 [22] 0009 ¶p020 [23] 0021 ¶p016 [24] 0055 ¶p083 [25] 0126 ¶p108

PREMARKET: Evaluating a device before it is sold

79 comments, 872 tags: 63 support, 635 conditional-minor, 123 conditional-major, 7 oppose, 44 neutral

Q7: Competency-based approach overall

47 comments, 121 tags: 31 support, 54 conditional-minor, 33 conditional-major, 3 neutral

Of 121 points, 31 support the approach as it stands, 54 ask for small changes, 33 for larger ones and 3 take no position; none opposes it. Supporters call benchmarking followed by clinical confirmation sound and appropriate, and a pragmatic alternative to exhaustive testing of open-ended outputs [1]. Several single out evaluating the final configured device rather than the foundation model as the paper's key principle, since prompts, retrieval, guardrails and interface change behavior [2].

The most common small change extends that principle: bind every result to one identified configuration of model version, prompts, retrieval, tools, guardrails and interface, and retest when it changes [3], or widen the unit of evaluation to the whole device pathway, workflow or sociotechnical system, including sensors, human oversight and monitoring [4]. For physical outputs, one commenter would evaluate the validated production system within a generative design envelope [5].

Several press the limits of the clinician analogy [6]. Credentialing is evidentially light only because of continuous accountability, licensure revocation and peer review, which a device lacks [7]; a model can be silently replaced and fail at scale in correlated ways [8]; licensing-style performance has not been shown to transfer to open practice, least of all for rare must-not-miss diagnoses [9]. Responses proposed: state that the analogy justifies the shape of evidence, not its quantity, and reject its use to loosen change control or spread accountability [10]; a prespecified route to suspend a device's authorization [11]; rebuilding psychometric properties such as case numbers and examiner reliability [12]; and validating deferral to a human as a testable property [13].

Several warn that competency is necessary but not sufficient and must not become a surrogate for clinical utility or outcomes, since passing a benchmark does not show improved care [14]; one would require clinical association, established in a research study, before any benchmarking [15]. On what competency means, one commenter would have it span knowledge, reasoning, recognizing danger and restraint from acting [16]; others ask that it be task-specific and consequence-sensitive because fluency is not judgment [17], be scoped like entrustment to a version, configuration, task, population and oversight level [18], and be tested on non-textbook presentations, pressure and unfamiliar patients [19].

Larger changes include a staged pipeline from supervised deployment to revalidation, with no device assumed permanently competent [20]; progressive expansion through predetermined evidence gates with rollback [21]; a supervised-operation stage whose entry evidence is lighter than independent use requires [22]; and making bounded inputs and outputs the primary expectation, with sponsors proposing competency evaluation justifying why bounding is unachievable [23]. One commenter would credential the deterministic gate and signing clinician around the model rather than the model itself [24]. The neutral entries describe verification infrastructure for sensor devices and illustrate bounding by intended use [25].

Sources. [1] 0003 ¶p011, ¶p030; 0006 ¶p029; 0028 ¶p023; 0043 ¶p016; 0052 ¶p069; 0087 ¶p022; 0088 ¶p021; 0091 ¶p035; 0093 ¶p016; 0100 ¶p040; 0102 ¶p005; 0106 ¶p013; 0114 ¶p003; 0117 ¶p004; 0123 ¶p014; 0124 ¶p002 [2] 0027 ¶p019; 0049 ¶p041; 0054 ¶p067; 0076 ¶p026; 0087 ¶p018; 0093 ¶p016; 0120 ¶p038; 0121 ¶p083; 0127 ¶p019 [3] 0005 ¶p264; 0035 ¶p053; 0076 ¶p012; 0088 ¶p024; 0115 ¶p018 [4] 0017 ¶p015; 0054 ¶p042; 0074 ¶p016; 0114 ¶p007; 0120 ¶p160 [5] 0053 ¶p008 [6] 0013 ¶p048; 0020 ¶p032; 0052 ¶p072; 0076 ¶p046; 0123 ¶p017; 0126 ¶p124 [7] 0013 ¶p048; 0020 ¶p032; 0076 ¶p012 [8] 0020 ¶p036; 0076 ¶p012

[9] 0126 ¶p124, ¶p127 [10] 0013 ¶p051; 0076 ¶p051 [11] 0076 ¶p048 [12] 0123 ¶p017 [13] 0126 ¶p128 [14] 0020 ¶p030; 0049 ¶p043; 0055 ¶p086; 0094 ¶p019; 0107 ¶p018 [15] 0113 ¶p019 [16] 0025 ¶p147 [17] 0056 ¶p004 [18] 0114 ¶p007; 0123 ¶p019 [19] 0058 ¶p054; 0118 ¶p026 [20] 0045 ¶p036 [21] 0052 ¶p079 [22] 0120 ¶p057 [23] 0126 ¶p031, ¶p139 [24] 0103 ¶p014 [25] 0074 ¶p036; 0126 ¶p369

Q8: Using the risk framework to scale premarket evidence

22 comments, 34 tags: 1 support, 23 conditional-minor, 9 conditional-major, 1 oppose

Of 34 points, 1 supports the paper, 23 ask for small changes, 9 for larger ones and 1 opposes. The opposing point says hazard analysis and the effectiveness of risk controls, not the two axes, should set the evidence [1]; the supporting one pairs rare autonomous diagnosis with heightened competency evidence and surveillance [2].

The most common request is a published mapping from risk tier or grid position to minimum benchmarking elements, confirmation tier, third-party role and monitoring, or to the regulatory pathway, so sponsors can plan [3]. One calls the approach without such a map case-by-case review under a new name [4]. Several sketch the gradient: benchmarking alone for low-risk informational functions, clinical confirmation such as shadow deployment as risk rises, and prospective, possibly randomized, evidence with intensive surveillance for autonomous high-consequence action [5].

Refinements: scale continuously rather than in rigid bins [6]; base evidence on residual risk after the actual safeguards, not on the model's latent capability [7]; validate any safeguard, such as human review, that lowers the burden [8]; raise assurance where system state lies outside the sponsor's control [9]; and count omission, delay and autonomy among the risks [10].

The larger changes pull in opposite directions. Two would lighten the middle of the grid [11]: benchmarking alone should presumptively suffice for lower and middle bands [12], and bounded functions should earn a lighter burden [13]. Three would tighten [14]: hard gates so a low average position cannot reduce evidence [15]; prospective scrutiny for all action-directing or action-taking devices regardless of consequence [16]; and clinical association established first, with both validations for every device [17].

Sources. [1] 0055 ¶p107 [2] 0121 ¶p052 [3] 0003 ¶p032; 0013 ¶p012, ¶p053; 0051 ¶p016; 0067 ¶p017; 0093 ¶p017 [4] 0013 ¶p053 [5] 0020 ¶p063; 0035 ¶p121; 0045 ¶p039; 0056 ¶p032; 0113 ¶p035; 0118 ¶p028 [6] 0049 ¶p045 [7] 0072 ¶p007; 0120 ¶p164 [8] 0052 ¶p070 [9] 0049 ¶p029 [10] 0025 ¶p331 [11] 0094 ¶p042; 0126 ¶p133 [12] 0094 ¶p042 [13] 0126 ¶p133 [14] 0005 ¶p310; 0107 ¶p016; 0113 ¶p021 [15] 0005 ¶p310 [16] 0107 ¶p016 [17] 0113 ¶p021

Q9: Benchmarking structure and methods

54 comments, 242 tags: 15 support, 184 conditional-minor, 24 conditional-major, 19 neutral

Of 242 points, 15 support the elements as they stand, 184 ask for small changes, 24 for larger ones and 19 take no position; none opposes. Supporters call the ten elements sound, sophisticated and well designed [1], and single out adjudicator independence extended to language-model adjudicators [2], both directions of escalation, refusal and deferral counted as failures [3], the escalation and deferral competencies themselves [4], and variation in safety-critical behavior treated as failure [5]. The changes are of two kinds: elements to add, and methods for scoring them.

Proposed additions cluster. Evidence integrity: separate measurement of citation fidelity, patient-specific facts, faithfulness to sources and fabricated specificity [6], since a knowledgeable output can still invent a patient's facts [7]; and an input-integrity element, because no element asks whether the device received any input, so fluent text over a dead stream passes [8]. Security and resilience: prompt

injection, including through retrieved documents, as a distinct element tied to cybersecurity expectations [9], applied also to non-agentic retrieval devices [10]; and tested behavior when dependencies or connectivity fail [11], with escalation subsystems failing closed [12]. Human factors as a standalone pillar linked to existing usability frameworks [13]. User-conditioned drift: sensitivity to a speaker's claimed seniority, persistent memory and sycophantic validation of a user's unsafe plan [14]; relatedly, a model's stated uncertainty is itself generated, so deferral should be tested adversarially and generated confidence told from computed [15]. Clinical competencies: latent clinical need in administrative requests [16], omission and harmful affirmation as failure types [17], refusing, escalating and recovering after failure [18], atypical presentations and chronic-disease reasoning [19], specialty modules [20], image grounding for radiology models [21], and therapeutic calibration for exposure-therapy devices, which the safety elements would otherwise fail [22].

On method, several want high-consequence failures and subgroups given their own pass-or-fail floors so averages cannot hide them [23], naming pediatric age bands, race, language and disability as strata [24]; one cautions that parity is not equity and would test usability for each group [25]. Reproducibility should be reported as a distribution under disclosed decoding settings, demonstrated behaviorally on action-relevant fields [26], and results should state where in a session they were sampled [27]. Language-model adjudicators should come from a different model family, with lineage disclosed and agreement with humans shown [28]. Other proposals: define the test article and configuration [29]; prespecify perturbations that must and must not change output [30]; emit claims in structured form for direct scoring [31]; credit architectural verification where deterministic logic governs safety behavior [32]; and ground safety-critical functions in curated evidence [33].

Few external standards are named: ISO/IEC 42001 and 23894, NIST's AI risk framework and AI reporting guidelines [34], usability and cybersecurity standards [35], and a machine-checkable dataset documentation layer [36]; two commenters offer their own open instruments [37], and one asks FDA to survey the field itself [38].

Sources. [1] 0020 ¶p015; 0067 ¶p018; 0078 ¶p017; 0091 ¶p012 [2] 0076 ¶p027; 0126 ¶p118; 0127 ¶p022 [3] 0076 ¶p006, ¶p028 [4] 0046 ¶p018 [5] 0126 ¶p104; 0127 ¶p023 [6] 0013 ¶p058; 0054 ¶p071; 0093 ¶p018; 0120 ¶p174; 0126 ¶p106 [7] 0054 ¶p071; 0120 ¶p174 [8] 0062 ¶p048; 0064 ¶p038; 0078 ¶p046 [9] 0013 ¶p057; 0018 ¶p013; 0032 ¶p057; 0102 ¶p008; 0117 ¶p007 [10] 0078 ¶p051 [11] 0032 ¶p051; 0111 ¶p050 [12] 0098 ¶p030 [13] 0018 ¶p017; 0024 ¶p007; 0067 ¶p023; 0094 ¶p045 [14] 0058 ¶p030; 0098 ¶p034; 0123 ¶p026 [15] 0076 ¶p050; 0127 ¶p033 [16] 0025 ¶p152 [17] 0021 ¶p009 [18] 0114 ¶p009 [19] 0118 ¶p030, ¶p032 [20] 0003 ¶p034 [21] 0120 ¶p171 [22] 0129 ¶p010 [23] 0005 ¶p296; 0021 ¶p010; 0025 ¶p159; 0028 ¶p024; 0052 ¶p081 [24] 0020 ¶p060; 0051 ¶p017; 0118 ¶p031; 0121 ¶p085 [25] 0059 ¶p029 [26] 0076 ¶p054; 0088 ¶p032; 0126 ¶p205 [27] 0078 ¶p033 [28] 0076 ¶p056; 0098 ¶p037; 0123 ¶p052 [29] 0067 ¶p021 [30] 0123 ¶p032 [31] 0126 ¶p151 [32] 0035 ¶p055; 0100 ¶p042; 0111 ¶p021 [33] 0025 ¶p143; 0121 ¶p045 [34] 0093 ¶p018 [35] 0018 ¶p013; 0067 ¶p023 [36] 0085 ¶p027 [37] 0078 ¶p008; 0091 ¶p035 [38] 0013 ¶p061

Q10: Benchmark validity and sponsor-developed benchmarks

40 comments, 91 tags: 3 support, 82 conditional-minor, 3 conditional-major, 3 neutral

Of 91 points, 3 support the paper, 82 ask for small changes, 3 for larger ones and 3 take no position; none opposes. Supporters value the paper's candor about contamination and saturation and say it is right to raise construct validity alongside sponsor benchmarks [1]. Several say public benchmarks are screens, not proof, and should not gate authorization alone [2].

On construct validity, several would require a documented argument for each benchmark stating what real-world behavior it predicts and the limits of that link [3]; several more want the link shown empirically, by correlation between benchmark results and independently adjudicated real-world or clinical-confirmation performance of the same device [4]. Others treat the benchmark as an instrument to qualify

first: a validity argument on the five sources of evidence in the educational testing standards [5]; a qualification package covering an independently fixed reference standard, adjudicator calibration, verification of the configuration run, a disagreement rule and versioned provenance [6]; a documented reproducibility floor, since saturation arrives when device spread falls below benchmark noise [7]; machine-checkable documentation of the dataset itself [8]; and fitness for purpose over dataset size [9]. One larger change holds that clinical association must be established before any benchmarking [10].

For contamination, proposals include a dated training-cutoff attestation from the model provider with temporally held-out data [11], provenance dating and canary items [12], duplicate screening and institutional holdouts [13], treating public adversarial assets as exposed [14], and disclosure of mitigations [15]. Many want sequestered or rotating test sets, held or scored by a party independent of sponsor and model developer [16]; the other larger change would make this a national repository with blinded third-party scoring [17].

On sponsor-developed benchmarks, several would permit them as unavoidable for narrow intended uses [18], with safeguards: locked and submitted before evaluation, attested unused in development, and partly escrowed [19]; a prespecified construction plan [20]; independent review of protocol, cases and rubric [21]; full disclosure [22]; custodian-held items with independent scoring [23]; and pass-fail bounds committed by hash so a moved bar is detectable [24]. One says such benchmarks can have higher validity than public assets built for other purposes [25]. Three would have consensus or specialty item banks developed through public-private processes or professional societies [26]. Benchmark content should also resemble practice: messy, minimizing patients and premature closure [27], premise-aware items with human adjudication [28], and conversational failures that single-turn scores mask [29]; one notes that lay users with a model found the condition far less often than the model alone [30].

Sources. [1] 0076 ¶p029; 0091 ¶p013 [2] 0013 ¶p066; 0017 ¶p027; 0045 ¶p048; 0048 ¶p005; 0087 ¶p023; 0107 ¶p012; 0114 ¶p017; 0115 ¶p022; 0120 ¶p180 [3] 0013 ¶p064; 0035 ¶p060; 0049 ¶p050; 0115 ¶p022; 0116 ¶p049 [4] 0028 ¶p024; 0093 ¶p019; 0100 ¶p045; 0113 ¶p024; 0118 ¶p034; 0123 ¶p040 [5] 0123 ¶p033 [6] 0091 ¶p024, ¶p025, ¶p028, ¶p029, ¶p030, ¶p031 [7] 0095 ¶p034 [8] 0085 ¶p025 [9] 0074 ¶p026 [10] 0113 ¶p023 [11] 0013 ¶p063 [12] 0067 ¶p026 [13] 0120 ¶p179 [14] 0027 ¶p075 [15] 0106 ¶p015 [16] 0003 ¶p037; 0005 ¶p254; 0013 ¶p067; 0045 ¶p049; 0051 ¶p018; 0056 ¶p014; 0067 ¶p026; 0093 ¶p019; 0100 ¶p046; 0102 ¶p010; 0116 ¶p076; 0117 ¶p010; 0120 ¶p179; 0123 ¶p042 [17] 0003 ¶p037 [18] 0013 ¶p065; 0067 ¶p026; 0082 ¶p031; 0093 ¶p019; 0120 ¶p181; 0123 ¶p042 [19] 0013 ¶p065 [20] 0067 ¶p026 [21] 0056 ¶p014; 0113 ¶p024; 0118 ¶p034 [22] 0049 ¶p050; 0082 ¶p031; 0120 ¶p181 [23] 0123 ¶p042 [24] 0131 ¶p069 [25] 0100 ¶p047 [26] 0028 ¶p024; 0118 ¶p034; 0123 ¶p042 [27] 0025 ¶p179 [28] 0082 ¶p030 [29] 0088 ¶p023 [30] 0123 ¶p035

Q11: Choosing a clinical confirmation approach

30 comments, 52 tags: 4 support, 37 conditional-minor, 8 conditional-major, 1 oppose, 2 neutral

Of 52 points, 4 support the paper, 37 ask for small changes, 8 for larger ones, 1 opposes and 2 take no position. Supporters call the ladder sound and welcome not requiring a prospective study in every case [1].

Many would have the rung follow risk: retrospective evaluation or shadow deployment for low-consequence or supervised functions, prospective study for autonomous or high-consequence ones [2]. Three ask for published presumptive tiers or a matrix keyed to the two-axis position, so sponsors know the default in advance [3], warning that trials for every function would consolidate the field to incumbents [4]. Two larger changes go further [5]: set the rung by the published risk mapping, not sponsor justification [6]; climb five tiers from retrospective to full deployment in order [7]. Others would let residual uncertainty after benchmarking drive the choice, treat the methods as complementary rather than a fixed hierarchy, and accept combinations [8]; one adds that a lighter approach justified by benchmark strength needs the benchmark itself qualified [9]. Two favor shadow deployment as an

underused middle tier that exposes real inputs without patient exposure [10].

On real-world inputs, commenters would characterize the anticipated distribution in advance, label evidence with its population and setting, and test case mixes that differ from training data [11]. Refinements include blinded adjudication, with the reference recorded before the output is seen, and task-qualified adjudicators [12]; temporal fidelity in retrospective tests [13]; physical testing for implant-generating functions [14]; longitudinal databanks as a rung [15]; and stress tests, varied settings, human factors and staged surgical introduction as added approaches [16].

The remaining larger changes pull the other way: prospective studies as principal evidence for devices directly influencing care, with outcome-based non-inferiority where outcomes are discrete [17], and clinical confirmation required for every device, never benchmarking alone [18]. The opposing point says not to cut corners and to run a trial openly [19].

Sources. [1] 0009 ¶p023; 0020 ¶p017; 0087 ¶p024; 0091 ¶p060 [2] 0003 ¶p039; 0005 ¶p314; 0006 ¶p024; 0017 ¶p028; 0025 ¶p335; 0049 ¶p052; 0055 ¶p156; 0093 ¶p020; 0118 ¶p035; 0120 ¶p187 [3] 0006 ¶p024; 0028 ¶p024; 0094 ¶p048 [4] 0003 ¶p039; 0006 ¶p024; 0094 ¶p048 [5] 0013 ¶p069; 0045 ¶p057 [6] 0013 ¶p069 [7] 0045 ¶p057 [8] 0028 ¶p015; 0030 ¶p009; 0052 ¶p085 [9] 0091 ¶p062 [10] 0013 ¶p070; 0118 ¶p036 [11] 0005 ¶p255; 0013 ¶p071; 0058 ¶p052; 0093 ¶p020 [12] 0009 ¶p025, ¶p026 [13] 0052 ¶p086 [14] 0053 ¶p008 [15] 0094 ¶p047 [16] 0017 ¶p029; 0038 ¶p027 [17] 0107 ¶p011, ¶p014 [18] 0113 ¶p025 [19] 0019 ¶p028

V-RWD: Real-world and ex-US data in clinical confirmation (no numbered question)

2 comments, 2 tags: 2 conditional-minor

Both points accept the use of real-world and non-US data in clinical confirmation and ask for small additions. One commenter would require a formal transportability assessment: characterizing case-mix, practice-pattern, coding and data-capture differences between source and target settings, with quantitative adjustment such as reweighting where differences are material, so that evidence from outside the United States is "neither privileged nor discounted" [1]. The other supports such data as the paper proposes but notes their limits: additional real-world cases arrive in proportion to how often each combination occurs, so thinly sampled strata stay thin; data from other health systems bring differences in population, practice and equipment whose relevance must be established; and because site, equipment, protocol and population tend to change together, the data cannot always show why performance falls, as in a cited study where models could identify the source hospital system almost perfectly [2].

Sources. [1] 0051 ¶p019 [2] 0126 ¶p261

Q12: Statistical rigor and combining evidence

22 comments, 47 tags: 41 conditional-minor, 2 conditional-major, 1 oppose, 3 neutral

Of 47 points, none supports the paper: 41 ask for small changes, 2 for larger ones, 1 opposes and 3 take no position. Several advise against pooling benchmarking and clinical confirmation into one estimate, since they sample different frames; any combination should be prespecified and justified [1]; the opposing point would not permit it at all, with confirmation governing acceptance and benchmarking as support [2]. Several likewise want synthetic and real-world results reported separately by default, combined only once transportability or a shared construct is shown under a prespecified method [3]. One larger change would let a prespecified feasibility study with validated outcomes serve small sponsors as both confirmation and benchmarking evidence [4].

The second theme is non-determinism. Several would measure variation over repeated runs at fixed

input [5]. One would report a per-endpoint reproducibility floor as the null, treating differences inside it as uninformative, citing a study where an agent's action changed in 8.7 percent of identical re-runs [6]; another would resample the unchanged configuration against itself before claiming change [7]. Related asks: acceptance judged on a lower confidence bound [8]; sampling settings under change control [9]; the case rather than the run as the unit of inference [10], clustering handled [11], and variance components used to set numbers of cases, runs and adjudicators [12]; and zero-failure bounds, under which bounding a rate below one percent needs about three hundred failure-free cases [13].

Other proposals: prespecify estimands, strata, intervals, multiplicity, subgroup precision and worst-case performance [14]; size samples by the precision risk-based claims need [15]; separate prevalence-representative evidence from risk-enriched coverage of rare hazards [16]; demand clinical as well as statistical significance [17]; and choose the outcome measure with care, since rigor cannot rescue task completion as the endpoint [18]. The other larger change requires clinical association before any sample-size question [19].

Sources. [1] 0005 ¶p300; 0051 ¶p020; 0055 ¶p171; 0091 ¶p065; 0093 ¶p021 [2] 0013 ¶p077 [3] 0005 ¶p315; 0049 ¶p053; 0055 ¶p170; 0120 ¶p192; 0126 ¶p273 [4] 0129 ¶p012 [5] 0013 ¶p073; 0035 ¶p054; 0055 ¶p162; 0120 ¶p190; 0123 ¶p043 [6] 0095 ¶p005; ¶p031 [7] 0091 ¶p033 [8] 0013 ¶p074 [9] 0013 ¶p076 [10] 0123 ¶p043 [11] 0055 ¶p161; 0120 ¶p191 [12] 0123 ¶p044 [13] 0123 ¶p046 [14] 0005 ¶p315; 0020 ¶p046; 0120 ¶p189 [15] 0052 ¶p088 [16] 0035 ¶p054; 0052 ¶p087 [17] 0045 ¶p060 [18] 0025 ¶p373 [19] 0113 ¶p026

Q13: Suitability and safeguards for synthetic data

26 comments, 49 tags: 2 support, 45 conditional-minor, 1 conditional-major, 1 neutral

Nearly every point adds conditions: of 49, 45 ask small changes, 2 support, 1 asks a larger change and 1 takes no position. Many commenters agree that synthetic data should supplement real data, not replace it [1]. Many name the same good uses: rare events, stress and adversarial tests, and controlled variation of one factor [2]. Poor uses: establishing performance rates or prevalence [3]; subgroup claims [4]; and communication, human behaviour and outcomes [5]. One commenter finds synthetic patients "too clinically tidy" and would model incomplete disclosure, distress and proxy questions [6].

Against same-class circularity, several safeguards recur: a generator of independent or different lineage [7]; disclosure of the generator, its method and its relation to the device [8]; validation against real distributions or confirmation of findings on real data [9]; labels or adjudication independent of the generating model [10]; and separate reporting of synthetic and real results [11]. Single suggestions: required out-of-distribution validation [12]; a stated minimum detectable difference for null subgroup results [13]; and proceeding as if no validated method exists for showing a synthetic population matches a real one [14]. The larger change would make a real-data fairness assessment a condition of approval for high-risk devices [15].

Both supporting points come from one commenter, who calls validated synthetic data the most direct way to strengthen generalizability evidence [16] and a complement to real-world evidence [17]; it would recognise validated synthetic benchmarks in submissions, PCCP verification and postmarket re-benchmarking once realism, perceptible findings and agreement with real-world results are shown [18]. Another treats simulated physiological signals as a complement to real recordings, with a simulator's limits stated [19]. The neutral point says limited studies give limited data for groups hard to include [20].

Sources. [1] 0003 ¶p042; 0017 ¶p033; 0020 ¶p050; 0025 ¶p348; 0035 ¶p062; 0045 ¶p062; 0049 ¶p056; 0052 ¶p089; 0055 ¶p179; 0076 ¶p055; 0113 ¶p027; 0120 ¶p195; 0126 ¶p276 [2] 0013 ¶p081; 0025 ¶p340; 0045 ¶p063; 0048 ¶p006; 0049 ¶p055; 0055 ¶p173; ¶p176; 0091 ¶p063; 0093 ¶p022; 0113 ¶p027; 0115 ¶p023; 0120 ¶p194; 0126 ¶p264 [3] 0013 ¶p082; 0048 ¶p006 [4] 0013 ¶p083; 0076 ¶p055; 0093 ¶p022; 0113 ¶p027; 0118 ¶p040 [5] 0025 ¶p344; 0055 ¶p179; 0126 ¶p271 [6] 0025 ¶p341, ¶p346, ¶p347 [7]

0005 ¶p260; 0013 ¶p083; 0045 ¶p063; 0049 ¶p056; 0055 ¶p186; 0076 ¶p055; 0093 ¶p022; 0126 ¶p274 [8] 0013 ¶p084; 0028 ¶p016; 0085 ¶p038; 0115 ¶p023; 0120 ¶p197 [9] 0028 ¶p016; 0049 ¶p056; 0055 ¶p187; 0076 ¶p055; 0118 ¶p040; 0120 ¶p197 [10] 0049 ¶p056; 0091 ¶p066; 0093 ¶p022; 0120 ¶p197 [11] 0017 ¶p033; 0048 ¶p006; 0052 ¶p089; 0076 ¶p055; 0115 ¶p023 [12] 0003 ¶p042 [13] 0095 ¶p008 [14] 0013 ¶p085 [15] 0113 ¶p027 [16] 0126 ¶p258 [17] 0126 ¶p276 [18] 0126 ¶p272, ¶p293 [19] 0074 ¶p019, ¶p021 [20] 0019 ¶p032

Q14: Comparators and acceptance criteria for open-ended outputs

36 comments, 98 tags: 1 support, 58 conditional-minor, 33 conditional-major, 1 oppose, 5 neutral

Of 98 points, 58 ask small changes and 33 larger ones; 1 supports, 1 opposes and 5 take no position. The supporting point agrees that no single comparator suits all devices [1].

Several reject the median clinician in practice as the primary standard because it normalizes existing gaps in care [2]. Their alternatives: prespecified acceptance criteria derived from the consequence of a wrong decision [3]; an evidence-based clinical standard or practice guidelines [4]; the standard of care as a floor with justified non-inferiority margins [5]; or the standard of care for safety and the realistic alternative for benefit claims [6]. Others would default to the median clinician, with consensus panels as a ceiling [7], let sponsors justify either standard [8], or fix a median from a documented sample to set an absolute threshold [9]. Objective ground truth, such as pathology or guidelines, should be used where it exists [10].

Several want specialists to set the reference for specialist tasks or high-stakes decisions [11], three of them even when the user is a generalist [12]; others would match the comparator to the labelled user and task rather than a job title [13]. One calls a specialist panel the wrong comparator for patient-facing information that displaces nothing, proposing the median generalist or a "usual information environment" [14]. Several treat the clinician panel as a measurement instrument whose inter-rater reliability must be reported [15]; one says a stable human reference standard must be demonstrated, not assumed [16]. The opposing point doubts clinicians would take part for fear of backlash [17].

Many would make human-AI team performance the basis of evaluation where a clinician reviews the output [18], several of them reserving device-alone performance for autonomous use [19]. Several still want device-alone results reported [20], and several insist team evaluation measure automation bias or whether users catch wrong outputs [21]. One would fix the output's workflow position in the intended use, since ordering changes team performance [22]; another cites evidence that adding a reviewer does not reliably raise accuracy [23].

Several larger changes concern acceptance criteria: safety-critical failures need separate, non-compensable floors that an average cannot offset [24]. Others want multidimensional criteria, covering such things as omissions, calibration, escalation and consistency [25], stricter than for one practitioner because errors scale [26], and the same for generative and conventional devices [27]. One would score the clinical claims in open-ended output against a reference standard with conventional sensitivity and specificity [28]; another sets provisional authorization at half the specialty baseline and independent authorization at parity [29]. The neutral points cite an autonomous screener whose labelled review step cannot catch missed cases [30] and an experiment in which patient trust fell when AI output was seen concurrently [31].

Sources. [1] 0126 ¶p174 [2] 0003 ¶p043; 0013 ¶p088; 0025 ¶p184; 0056 ¶p015; 0076 ¶p059 [3] 0013 ¶p090 [4] 0025 ¶p185; 0118 ¶p044 [5] 0076 ¶p059 [6] 0123 ¶p050 [7] 0100 ¶p049 [8] 0028 ¶p024; 0093 ¶p023 [9] 0126 ¶p181 [10] 0055 ¶p188; 0113 ¶p029 [11] 0003 ¶p044; 0055 ¶p190; 0056 ¶p009; 0076 ¶p059; 0113 ¶p029; 0118 ¶p043; 0123 ¶p050; 0126 ¶p177 [12] 0056 ¶p009; 0076 ¶p059; 0123 ¶p050 [13] 0013 ¶p093; 0045 ¶p029; 0049 ¶p057 [14] 0006 ¶p025 [15] 0009 ¶p033; 0013 ¶p092; 0100 ¶p049; 0106 ¶p013; 0123 ¶p049 [16] 0009 ¶p032 [17] 0019 ¶p034 [18] 0013 ¶p094; 0017 ¶p030; 0028 ¶p024; 0035 ¶p062; 0045 ¶p066; 0046 ¶p018; 0052 ¶p095; 0054 ¶p076; 0056 ¶p017; 0070 ¶p009; 0087 ¶p027; 0093 ¶p023; 0094 ¶p050; 0100 ¶p051; 0118 ¶p042;

0120 ¶p201; 0123 ¶p053 [19] 0035 ¶p062; 0045 ¶p066; 0093 ¶p023; 0094 ¶p050; 0100 ¶p051; 0118 ¶p042 [20] 0013 ¶p094; 0028 ¶p024; 0046 ¶p018; 0055 ¶p192; 0056 ¶p017; 0120 ¶p202; 0123 ¶p053 [21] 0013 ¶p095; 0017 ¶p031; 0054 ¶p006; 0056 ¶p017; 0058 ¶p036; 0076 ¶p059; 0086 ¶p005; 0094 ¶p051; 0120 ¶p070; 0123 ¶p053 [22] 0070 ¶p013 [23] 0046 ¶p018 [24] 0052 ¶p081; 0055 ¶p087; 0056 ¶p014; 0064 ¶p030; 0114 ¶p013; 0123 ¶p051 [25] 0049 ¶p058; 0056 ¶p018; 0115 ¶p019; 0120 ¶p073 [26] 0076 ¶p049 [27] 0126 ¶p185 [28] 0126 ¶p144 [29] 0045 ¶p110 [30] 0070 ¶p011 [31] 0070 ¶p012

Q15: Comparing with care in the device's absence

23 comments, 44 tags: 3 support, 32 conditional-minor, 4 conditional-major, 2 oppose, 3 neutral

Of 44 points, 3 support, 32 ask small changes, 4 larger ones, 2 oppose and 3 take no position. Most commenters endorse comparing a device with the care that would actually occur without it [1]. The usual reason is access: where care is scarce, the alternative is often no assessment, delay or self-directed searching [2].

Several want the counterfactual characterized from data, not assumed [3]. Others want it prespecified and justified [4]; carried into labelling so a low-bar authorization cannot spread to richer settings [5]; kept a supplement to a specialist panel [6]; and paired with results such as time to care and errors prevented [7]. One considers no intervention the only reliable comparator [8]; another would make the unaided user primary where the intended use is "when no professional is available" [9].

The larger changes keep an absolute standard alongside: a safety floor so a poor baseline cannot institutionalize substandard care [10]; an evidence-based ideal [11]; absent care weighed in benefit-risk only, with absolute acceptance thresholds [12]; and a comparator distinct from the reference standard, judged by non-inferiority [13]. One opposing point says absence changes a device's benefit, not the consequence of its errors [14]; the other calls unaided judgment and delayed review pointless [15].

One commenter, on communication devices for nonverbal users, says absence is the only comparator describing reality [16]: a fabricated utterance lands below zero, since no false record would otherwise exist [17]; the absence comparator is a floor test, not a replacement [18], required when no human performs the function, the user cannot contest output and errors leave an attributed record [19]. Its neutral points report words transcribed from dead audio [20] and take no position where a human alternative exists but is unavailable [21].

Sources. [1] 0003 ¶p044; 0005 ¶p318; 0006 ¶p025; 0009 ¶p035; 0013 ¶p096; 0020 ¶p041; 0045 ¶p066; 0049 ¶p060; 0051 ¶p021; 0055 ¶p193; 0082 ¶p032; 0093 ¶p024; 0100 ¶p050; 0113 ¶p030; 0118 ¶p046; 0120 ¶p205 [2] 0003 ¶p044; 0013 ¶p096; 0100 ¶p050; 0118 ¶p046 [3] 0009 ¶p040; 0013 ¶p097; 0051 ¶p021; 0093 ¶p024; 0100 ¶p050; 0120 ¶p209 [4] 0049 ¶p061; 0113 ¶p030 [5] 0013 ¶p097 [6] 0118 ¶p046 [7] 0055 ¶p194; 0120 ¶p211 [8] 0043 ¶p024 [9] 0082 ¶p008 [10] 0017 ¶p032 [11] 0025 ¶p204 [12] 0126 ¶p182 [13] 0052 ¶p091, ¶p093 [14] 0126 ¶p180 [15] 0019 ¶p037 [16] 0064 ¶p009 [17] 0064 ¶p011 [18] 0064 ¶p029 [19] 0064 ¶p035 [20] 0064 ¶p021 [21] 0064 ¶p040

Q16: Role of independent third parties

35 comments, 63 tags: 2 support, 53 conditional-minor, 6 conditional-major, 1 oppose, 1 neutral

Of 63 points, 53 ask small changes, 6 larger ones, 2 support, 1 opposes and 1 takes no position. Most commenters welcome qualified third parties in the roles the paper names, above all holding sequestered datasets and supplying independent adjudication [1]. One supporting point finds independence most valuable where sponsors pick their own benchmarks or another company supplies the model [2].

The small changes are mostly design safeguards. Many warn against a single gatekeeper or exclusive certification market, asking for multiple bodies, published criteria or scaled fees [3]. Several would keep third-party assessment voluntary, preserving a first-party or self-assessment route [4]. On independence,

several want financial ties disclosed or payment untied from results [5]; some want independence from the foundation model developer as well as the sponsor [6], two of them adding that an LLM adjudicator must not share the device's lineage [7]. One would bar a body that advised a sponsor from assessing its device [8]; another notes that institutional independence is not operational independence when evaluator and device share a reference standard or model family [9]. Several would keep manufacturers accountable or acceptance decisions inside CDRH [10].

Alternatives to certifying devices: qualify benchmark suites or tools through MDDT rather than designate providers [11]; open methods and reproducible sponsor testing, with independence targeted at adjudication and sequestered safety sets [12]; and confidence resting on how an evaluation is run, prespecified, automatically scored and auditable [13]. Two would make professional societies eligible for specialty datasets and adjudicators [14]; another would have societies, academia and labs profile devices against standard competencies [15]. One notes third-party capacity for generative devices does not yet exist at scale [16].

The larger changes would make independent testing mandatory: before market with recurring audits [17], for medium- and high-risk devices [18], or for the safety-critical elements where self-assessment cannot resolve the conflict of interest [19]; another would recognise CLIA and CAP accreditation as satisfying these elements for laboratory-embedded functions [20]. The opposing point calls a third-party requirement inadvisable because it invites gaming and profit centres [21]. The neutral point offers a lab's methods for comparison with the paper's elements [22].

Sources. [1] 0003 ¶p045; 0005 ¶p319; 0006 ¶p028; 0013 ¶p098; 0025 ¶p210; 0028 ¶p024; 0029 ¶p015; 0035 ¶p064; 0045 ¶p069; 0049 ¶p062; 0055 ¶p195; 0056 ¶p019; 0076 ¶p057; 0091 ¶p067; 0093 ¶p025; 0100 ¶p052; 0115 ¶p024; 0116 ¶p048; 0118 ¶p048; 0120 ¶p214 [2] 0029 ¶p016 [3] 0003 ¶p046; 0005 ¶p319; 0006 ¶p028; 0013 ¶p100; 0035 ¶p064; 0045 ¶p069; 0049 ¶p063; 0055 ¶p203; 0056 ¶p021; 0071 ¶p015; 0076 ¶p057; 0093 ¶p025; 0100 ¶p053; 0115 ¶p024; 0116 ¶p077; 0118 ¶p048; 0120 ¶p216 [4] 0013 ¶p103; 0094 ¶p055; 0100 ¶p054; 0111 ¶p040; 0120 ¶p216; 0126 ¶p291 [5] 0013 ¶p102; 0025 ¶p211; 0035 ¶p064; 0055 ¶p202; 0056 ¶p020; 0115 ¶p024 [6] 0071 ¶p014; 0093 ¶p025; 0100 ¶p055; 0113 ¶p031 [7] 0071 ¶p014; 0100 ¶p055 [8] 0013 ¶p101 [9] 0104 ¶p014 [10] 0049 ¶p063; 0055 ¶p204; 0093 ¶p025; 0120 ¶p217; 0126 ¶p291 [11] 0067 ¶p028; 0126 ¶p291, ¶p294 [12] 0091 ¶p067 [13] 0126 ¶p290 [14] 0116 ¶p050; 0118 ¶p048 [15] 0114 ¶p027 [16] 0013 ¶p104 [17] 0047 ¶p005 [18] 0113 ¶p031 [19] 0071 ¶p012 [20] 0094 ¶p054 [21] 0043 ¶p025 [22] 0021 ¶p025

Q17: Other model architectures

16 comments, 29 tags: 1 support, 24 conditional-minor, 4 neutral

Of 29 points, 24 ask small changes, 1 supports and 4 take no position. Most commenters say the competency approach carries over to other architectures [1], several stressing that it judges the deployed device's behaviour rather than its internals [2]. The small changes add architecture-specific tests. For multimodal devices: cross-modal consistency or grounding, such as the same case given as image and as text [3]; proof that the image drives the answer, by ablation and substitution [4] or by cases with no image, where one experiment found frontier models diagnosing anyway [5]; robustness to capture quality, equipment, lighting or skin tone [6]; each modality benchmarked separately with the integrated device shown not degraded [7]; and cases where normative colour codes diverge from the correct reading [8]. For world models: error over multiple predicted steps, counterfactuals and confirmation that predicted trajectories match real patients [9], with one asking for an explicit branch separating content generation from predictive fidelity before elements are chosen [10] and another pointing to computational-modelling credibility practice [11].

Cautions: thin benchmarks for these modalities mean clinical confirmation must carry more weight [12]; FDA should publish applicability notes per architecture [13] or companion multimodal guidance [14]; and conversational elements do not fit imaging measurement functions, which existing guidance should

keep governing [15]. The neutral points describe the no-image experiment and its caveats [16].

Sources. [1] 0005 ¶p321; 0045 ¶p071; 0049 ¶p064; 0055 ¶p108; 0093 ¶p026; 0108 ¶p001; 0113 ¶p033; 0118 ¶p050; 0120 ¶p219; 0123 ¶p054 [2] 0025 ¶p376; 0045 ¶p071; 0049 ¶p064; 0118 ¶p050 [3] 0049 ¶p064; 0055 ¶p109; 0113 ¶p033; 0120 ¶p220 [4] 0123 ¶p055 [5] 0108 ¶p003, ¶p010 [6] 0055 ¶p113; 0093 ¶p026; 0118 ¶p050; 0123 ¶p057 [7] 0003 ¶p047 [8] 0123 ¶p056 [9] 0055 ¶p114, ¶p115; 0113 ¶p033 [10] 0129 ¶p014 [11] 0093 ¶p026 [12] 0013 ¶p106 [13] 0093 ¶p026 [14] 0003 ¶p047 [15] 0126 ¶p168, ¶p172 [16] 0108 ¶p002, ¶p011

POSTMARKET: Oversight once a device is in use

92 comments, 793 tags: 21 support, 631 conditional-minor, 72 conditional-major, 9 oppose, 60 neutral

Q18: Trading premarket uncertainty for postmarket monitoring

43 comments, 122 tags: 4 support, 89 conditional-minor, 16 conditional-major, 9 oppose, 4 neutral

Of 122 points, 4 support the trade, 89 ask small changes, 16 larger ones, 9 oppose and 4 take no position. Supporters call the trade sound [1], strongly support it [2] or agree that more of the safety case will be carried after market [3]. The opposing points, from four commenters, hold that monitoring must supplement or verify premarket evidence, not replace it [4], one adding that reduced evidence transfers the discovery of uncertainty to patients [5]. One commenter would go the other way, making the trade the default outside the highest-risk band as least burdensome and more protective for devices that change after deployment [6].

Most commenters, whatever their stance, would exclude high-consequence, autonomous or irreversible functions [7]. Specific exclusions: permanent implants [8]; anaphylaxis guidance and autonomous dosing [9]; diagnoses for serious conditions without prior qualified review [10]; and, from one commenter, users who cannot self-report, offline operation and single-user long-duration deployment [11]. Many condition the trade on a detection interval shorter than the time to harm, or on reversibility or rollback [12]. Larger changes would have each function declare a reversibility class, irreversible ones being ineligible [13], or make reduced evidence contingent on automated runtime kill-switches [14].

Conditions on the monitoring program recur. Many want it prespecified before deployment, with metrics, thresholds, triggers, cadence and corrective actions [15], some adding that it be enforceable with automatic consequences such as suspension [16]. Several want detection capability demonstrated, not a program's existence: coverage documented, the failure classes it misses stated, a data path from output to observable outcome shown [17], with a larger change tying each reduction to a named failure mode and a detection demonstration [18]. Three want the monitored record independent of the device or of sponsor attestation [19]. Others want monitoring continuous from first deployment [20], operational at launch [21] or near real time for safety-critical behaviours [22]; a machine supervisor's reliability shown before any credit [23]; and deployers able to inventory, limit, log and stop the device [24]. Two draw on history: postmarket obligations are fulfilled less reliably than premarket ones [25], and drug-safety systems detect only prespecified harms and count timeliness rather than detection [26]. One lists prerequisites surveillance lacks for GenAI: a failure taxonomy, public thresholds, a patient reporting channel and public results [27]. The neutral points report within-session degradation that periodic re-benchmarking cannot see [28] and note the docket divides on what monitoring reads [29].

Sources. [1] 0003 ¶p049 [2] 0100 ¶p011 [3] 0130 ¶p003 [4] 0023 ¶p006; 0086 ¶p008; 0113 ¶p036; 0126 ¶p188 [5] 0126 ¶p188 [6] 0094 ¶p057, ¶p059 [7] 0005 ¶p322; 0013 ¶p115; 0017 ¶p005; 0020 ¶p054; 0024 ¶p009; 0025 ¶p218; 0028 ¶p025; 0035 ¶p066; 0045 ¶p075; 0049 ¶p068; 0051 ¶p022; 0053 ¶p009; 0054 ¶p080; 0055 ¶p214; 0076 ¶p035; 0090 ¶p015; 0093 ¶p027; 0094 ¶p061; 0100 ¶p057; 0105 ¶p026; 0115 ¶p025; 0116 ¶p025; 0118 ¶p052; 0120 ¶p228; 0126 ¶p194; 0131 ¶p027 [8] 0053 ¶p009 [9] 0118 ¶p052 [10] 0126 ¶p194 [11] 0077 ¶p046, ¶p048, ¶p052 [12] 0005 ¶p322; 0013 ¶p113; 0017 ¶p036; 0048 ¶p007; 0049 ¶p067; 0052 ¶p119; 0055 ¶p214; 0075 ¶p021; 0077 ¶p050; 0093 ¶p027; 0105 ¶p026; 0115 ¶p025; 0126 ¶p191 [13] 0131 ¶p027 [14] 0102 ¶p006; 0117 ¶p006 [15] 0003 ¶p050; 0006 ¶p026; 0013 ¶p111; 0028 ¶p025; 0035 ¶p066; 0045 ¶p075; 0075 ¶p020; 0090 ¶p012; 0094 ¶p061; 0099 ¶p025; 0100 ¶p057; 0120 ¶p116 [16] 0003 ¶p050; 0013 ¶p111; 0090 ¶p014 [17] 0008 ¶p013; 0013 ¶p112; 0033 ¶p014; 0090 ¶p015; 0093 ¶p027; 0099 ¶p024; 0116 ¶p025 [18] 0077 ¶p031 [19] 0077 ¶p035; 0090 ¶p013; 0127 ¶p039 [20] 0090 ¶p011 [21] 0116 ¶p024 [22] 0118 ¶p052 [23] 0111 ¶p038 [24] 0130 ¶p009 [25] 0013 ¶p110 [26] 0033 ¶p012 [27] 0076 ¶p036 [28] 0077 ¶p027 [29] 0127 ¶p038

Q19: Postmarket performance evaluation approaches

70 comments, 238 tags: 10 support, 183 conditional-minor, 17 conditional-major, 28 neutral

Of 238 points, 10 support, 183 ask small changes, 17 larger ones and 28 take no position. Many commenters find the three approaches reasonable or complementary [1]; one calls them layers of one program rather than a menu [2]. Two propose risk-tiered fixed cadences, from annual review for low risk to quarterly or monthly for high [3]; many more want reassessment event-triggered as well as periodic, not on a calendar alone [4]. The triggers named most often are a change to the model, including by a third-party developer, or to prompts, retrieval, guardrails or tools; a shift in the input population or a new site; complaint or adverse-event clusters; and breach of a prespecified threshold [5].

Many would monitor more than accuracy: clinically meaningful outcomes, near misses, overrides, automation bias and subgroup performance [6]; one would add outcome-linked surveillance in claims and EHR data as a fourth approach [7]. Two give input-population drift standing as its own signal, since aggregate accuracy can hold while a subgroup degrades [8]. One warns that representative sampling cannot show adversarial resistance, so safety elements need re-benchmarking against an adversarial battery on their own cadence [9]. Several insist monitoring link to prespecified actions such as investigation, rollback or suspension [10].

Many want records that trace each output to the model version, prompts and configuration that produced it, enough to reconstruct a consequential event [11]; two call such logs a precondition for any of the approaches [12]. Larger changes would mandate a reconstructable event record for higher-consequence devices [13] or a six-field chain-of-custody record per output with automated kill-switch tiers [14]. Several want detection independent of the device's own reporting [15]; one adds that absent input and dropped sessions must be logged like errors [16]. On clinician review, several would combine random with risk-weighted or stratified sampling, oversampling high-acuity and safety-critical interactions [17]; one would measure and report adjudicator agreement, with criteria written as decision rules [18]; another would reserve such review for higher-risk devices [19].

Blind spots raised: degradation within a session resets at its boundary, so periodic re-benchmarking never sees it, and the larger change is continuous within-session evaluation with an independent access record and session-end notification to a caregiver [20]; each rate's population must be declared [21]; the reproducibility floor sets the minimum detectable effect [22]; frozen on-device models cannot drift or phone home [23]; and MDR reportability of caught hallucinations is unresolved [24]. Other larger changes: mandatory participation in a qualified data system linking use to outcomes for action-directing functions [25], and a discontinuation plan in the authorization [26]. Neutral points include a review of 95 authorized PCCPs that found one monitoring cadence and one quantified trigger [27], a study of device reports showing passive reporting's limits [28], and one commenter's description of its own review architecture [29].

Sources. [1] 0013 ¶p116; 0020 ¶p018; 0029 ¶p020; 0049 ¶p069; 0052 ¶p121; 0058 ¶p039; 0071 ¶p018; 0087 ¶p029; 0090 ¶p021; 0093 ¶p028; 0100 ¶p058; 0116 ¶p043; 0118 ¶p054; 0120 ¶p230; 0126 ¶p197 [2] 0090 ¶p021 [3] 0003 ¶p052; 0113 ¶p037 [4] 0005 ¶p323; 0013 ¶p116; 0017 ¶p039; 0035 ¶p073; 0045 ¶p078; 0048 ¶p008; 0049 ¶p070; 0052 ¶p123; 0054 ¶p081; 0056 ¶p022; 0071 ¶p021; 0075 ¶p022; 0090 ¶p022; 0091 ¶p058; 0093 ¶p028; 0100 ¶p060; 0105 ¶p027; 0114 ¶p034; 0115 ¶p026; 0116 ¶p044; 0118 ¶p054; 0120 ¶p234; 0130 ¶p010 [5] 0013 ¶p117; 0035 ¶p074; 0054 ¶p092; 0056 ¶p023; 0090 ¶p022; 0093 ¶p028; 0115 ¶p027; 0120 ¶p234 [6] 0017 ¶p038; 0021 ¶p018; 0025 ¶p228; 0039 ¶p004; 0052 ¶p121; 0054 ¶p091; 0055 ¶p224; 0056 ¶p024; 0086 ¶p009; 0087 ¶p029; 0097 ¶p007; 0116 ¶p032; 0120 ¶p075; 0121 ¶p014 [7] 0051 ¶p023 [8] 0100 ¶p061; 0122 ¶p020 [9] 0027 ¶p045 [10] 0052 ¶p110; 0055 ¶p227; 0067 ¶p030; 0075 ¶p022; 0105 ¶p028; 0116 ¶p044; 0118 ¶p055; 0120 ¶p076 [11] 0047 ¶p006; 0050 ¶p009; 0052 ¶p141; 0087 ¶p043; 0092 ¶p015; 0098 ¶p043; 0105 ¶p041; 0115 ¶p032; 0116 ¶p031; 0120 ¶p080; 0130 ¶p010 [12] 0098 ¶p043; 0130 ¶p010 [13] 0049 ¶p073 [14] 0102 ¶p009; 0117 ¶p008 [15] 0008 ¶p015; 0063 ¶p022; 0078 ¶p088; 0103 ¶p015; 0104 ¶p019 [16] 0062 ¶p046 [17] 0055 ¶p225; 0056 ¶p023; 0092 ¶p005; 0100 ¶p059; 0116 ¶p043; 0118 ¶p054 [18] 0033 ¶p022, ¶p024 [19] 0028

¶p025 [20] 0063 ¶p016, ¶p031, ¶p041 [21] 0131 ¶p045 [22] 0095 ¶p041 [23] 0082 ¶p033 [24] 0013 ¶p129; 0033 ¶p038 [25] 0116 ¶p022 [26] 0021 ¶p017 [27] 0090 ¶p019 [28] 0116 ¶p012 [29] 0092 ¶p031

Q20: Machine-based supervisory agents

43 comments, 104 tags: 3 support, 88 conditional-minor, 7 conditional-major, 6 neutral

Of 104 points, 3 support, 88 ask small changes, 7 larger ones and 6 take no position: commenters accept supervisory agents but attach conditions. One supporter calls them likely the only approach that scales for high-volume, open-ended devices [1]; another points to an EU-certified autonomous breast screening device whose supervisory layer returns sites to full radiologist reading [2]. Many insist the agent triage, flag or escalate but never conclude: human review stays the final layer, and a machine must not be the sole supervisor of another machine [3]. One cautions that automation suits deterministic checks, not the clinical judgment that sample review exists for [4].

Correlated failure is raised most often: a supervisor built on the device's model family may share its blind spots, so its agreement cannot count as independent confirmation [5]. Several want a different model family or provider [6], though two add that a different model or vendor alone is not enough [7]. Others propose reporting failure-mode correlation and scaling credit to it [8], challenge sets built from known device failures [9], or a supervisor organizationally separate from the device or its manufacturer [10].

Many would validate the supervisor itself before crediting it [11]; several specify sensitivity and specificity for the failure modes it flags, measured against human adjudication and re-confirmed over time [12], or misses and false alarms characterized separately [13]. Two would treat it as a device software function or safety-relevant component [14]; one would make it pass the paper's safety benchmarking elements [15]. Several want humans to sample cleared as well as flagged cases [16]. Several say supervisor changes should trigger reassessment like device changes [17], two naming the prompt [18], and several ask for audited, tamper-evident or append-only logs [19]. Several would define the supervisor's role and authority, distinguishing observing, recommending and intervening, with limited permissions and fail-safe behavior when it is unavailable [20]; one suggests observation-only deployment before any authority to alter output [21].

One commenter's own supervisor nearly flagged a true, clinically vital statement as fabricated because its ground-truth definition was too narrow [22]; another recorded four occasions when a model acting as reviewer asserted what its evidence could not support, offered as a failure class, not a rate [23], and would require each finding to state its basis and timestamp first and carry an evidence class [24]. One cites judge-model instability to argue a supervisor cannot be credited for changes smaller than its own instability [25]; one finds the HL7 audit standard cannot express one agent acting for another [26].

The larger changes: one commenter would permit supervisory agents only where their output cannot be read downstream as human judgment, through four record properties including a model-assistance flag [27]; another would let the supervisor read every record but write nothing to the record of fact [28]; a third wants a required human-controlled kill switch [29]. One suggests qualifying supervisory methods through MDDT [30], another a separate docket for supervisor standards [31].

Sources. [1] 0090 ¶p024 [2] 0126 ¶p210 [3] 0013 ¶p124; 0025 ¶p237; 0052 ¶p132; 0076 ¶p016; 0086 ¶p009; 0093 ¶p029; 0100 ¶p062; 0102 ¶p010; 0113 ¶p038; 0117 ¶p010; 0118 ¶p057; 0120 ¶p237 [4] 0033 ¶p031 [5] 0013 ¶p120; 0017 ¶p040; 0032 ¶p101; 0035 ¶p076; 0052 ¶p131; 0055 ¶p235; 0056 ¶p030; 0076 ¶p063; 0090 ¶p026; 0093 ¶p029; 0095 ¶p044; 0104 ¶p015; 0106 ¶p017; 0116 ¶p047; 0118 ¶p057; 0120 ¶p239 [6] 0013 ¶p122; 0055 ¶p237; 0090 ¶p026; 0110 ¶p003 [7] 0075 ¶p044; 0115 ¶p034 [8] 0013 ¶p122 [9] 0090 ¶p026 [10] 0071 ¶p026; 0130 ¶p011 [11] 0003 ¶p055; 0013 ¶p123; 0033 ¶p030; 0045 ¶p081; 0049 ¶p079; 0052 ¶p130; 0055 ¶p229; 0075 ¶p044; 0090 ¶p025; 0093 ¶p029; 0115 ¶p037; 0116 ¶p047; 0120 ¶p238 [12] 0003 ¶p055; 0013 ¶p123; 0090

¶p025; 0120 ¶p238 [13] 0033 ¶p030; 0049 ¶p079; 0075 ¶p044; 0115 ¶p037 [14] 0115 ¶p041; 0126 ¶p211 [15] 0106 ¶p017 [16] 0055 ¶p239; 0075 ¶p045; 0093 ¶p029 [17] 0013 ¶p125; 0075 ¶p045; 0090 ¶p028; 0095 ¶p044; 0105 ¶p030 [18] 0090 ¶p028; 0095 ¶p044 [19] 0035 ¶p076; 0090 ¶p027; 0105 ¶p030; 0115 ¶p039 [20] 0049 ¶p077; 0075 ¶p042, ¶p046; 0105 ¶p030; 0115 ¶p038 [21] 0111 ¶p036 [22] 0111 ¶p033 [23] 0066 ¶p009, ¶p056 [24] 0066 ¶p049, ¶p053 [25] 0095 ¶p044 [26] 0042 ¶p007 [27] 0131 ¶p051, ¶p054 [28] 0127 ¶p040 [29] 0118 ¶p058 [30] 0090 ¶p029 [31] 0003 ¶p055

Q21: Shared ecosystem roles and accountability

44 comments, 121 tags: 2 support, 94 conditional-minor, 9 conditional-major, 16 neutral

Of 121 points, 2 support, 94 ask small changes, 9 larger ones and 16 take no position. Most commenters accept shared participation on one condition: the manufacturer keeps accountability, and others contribute without taking it over [1]. Formulations vary: information may come from anywhere but the duty to investigate stays put [2]; stakeholders detect and report, the manufacturer decides [3]; distribute evidence generation, not accountability [4]; write each role as an input the manufacturer must act on, not a duty satisfied in its place [5]. Several add that contracts or vendor relationships transfer work, not accountability [6], and several would let accountability follow what each party controls, knows or can see [7].

Many sketch the division. Institutions handle local validation, workflow, access, inventory, logs, reporting pathways or incident escalation [8]; clinicians report errors, near misses and discrepancies seen in use [9]; professional societies supply failure taxonomies, performance measures, reviewer qualifications or adjudicator calibration libraries [10]; standards bodies define formats for logs, versions and incidents, including attribution semantics [11]; model providers disclose changes [12]. One proposes that health plans supply monitoring samples and outcome linkage under contract [13]; others would add technology-management and cybersecurity staff [14], nurses and patients [15], or research-site operators [16]. To make roles operational, several want each party's duties written down [17], for instance a role matrix naming who monitors, pauses, investigates and reports [18] or who decides and who implements each response step [19], with data-sharing agreements [20], rehearsed handoffs [21], and durable records of what the AI said and who reviewed it [22], which one says lets others contribute signal without absorbing accountability [23].

Capacity is the counterweight. One commenter, in seven of the nine larger changes, says most health systems cannot inventory their software devices or record which model version produced an output, so the institutional role should be receiving and reporting, not evaluation, with FDA specifying what manufacturers must supply and weighing a heavier manufacturer burden [24]. Others agree institutions should not re-validate models [25], want minimum site capacity addressed through CMS infrastructure [26], say a small practice has nobody to watch the model [27], and would keep continuous surveillance off pathology laboratories [28] and uncompensated review off societies [29]. One asks that institutions configuring within the validated envelope not count as manufacturers [30]. Against this, several would keep local validation, or a proportionate deployment qualification, as a layer of its own distinct from the manufacturer's testing [31]. Several want clinicians protected from absorbing sole liability for device errors [32].

Pooled infrastructure: a cross-manufacturer sentinel network like the Vaccine Safety Datalink, the other larger change [33]; an FDA-supported confidential multi-site network [34]; qualified data systems governed with societies, independent of industry, with multiple operators and FDA-CMS coordination [35]. Several want low-friction reporting routes for clinicians, patients and caregivers [36]; one notes its users cannot complain at all [37]. One would suspend authorization where adequate monitoring and review cannot be done [38]. Neutral points offer state coverage law as a record of role drafting [39],

registry precedents [40], and a proposed AI nursing association [41].

Sources. [1] 0003 ¶p057; 0005 ¶p325; 0013 ¶p126; 0017 ¶p039; 0025 ¶p248; 0032 ¶p101; 0033 ¶p035; 0035 ¶p078; 0045 ¶p084; 0052 ¶p133; 0055 ¶p240; 0056 ¶p025; 0075 ¶p026; 0076 ¶p037; 0086 ¶p009; 0087 ¶p041; 0092 ¶p008; 0093 ¶p030; 0099 ¶p015; 0100 ¶p063; 0105 ¶p032; 0113 ¶p039; 0116 ¶p039; 0118 ¶p060; 0119 ¶p037; 0120 ¶p245; 0130 ¶p012 [2] 0013 ¶p126 [3] 0093 ¶p030 [4] 0005 ¶p325 [5] 0119 ¶p037 [6] 0025 ¶p259; 0032 ¶p101; 0076 ¶p037; 0099 ¶p016 [7] 0017 ¶p007; 0025 ¶p263; 0049 ¶p082; 0130 ¶p012 [8] 0017 ¶p051; 0033 ¶p034; 0055 ¶p242; 0056 ¶p025; 0087 ¶p041; 0092 ¶p011; 0113 ¶p039; 0119 ¶p039; 0120 ¶p243; 0130 ¶p012 [9] 0017 ¶p051; 0055 ¶p243; 0056 ¶p025; 0119 ¶p038; 0120 ¶p243 [10] 0003 ¶p057; 0033 ¶p033; 0055 ¶p244; 0056 ¶p025; 0092 ¶p120; 0119 ¶p040; 0120 ¶p243 [11] 0042 ¶p054; 0055 ¶p245; 0120 ¶p244 [12] 0003 ¶p057; 0120 ¶p119 [13] 0009 ¶p043 [14] 0032 ¶p042 [15] 0081 ¶p001 [16] 0112 ¶p014 [17] 0047 ¶p012; 0049 ¶p084; 0075 ¶p027; 0105 ¶p032 [18] 0105 ¶p032 [19] 0075 ¶p027 [20] 0035 ¶p078; 0052 ¶p134; 0120 ¶p245 [21] 0075 ¶p030 [22] 0025 ¶p256; 0046 ¶p005 [23] 0127 ¶p040 [24] 0050 ¶p007, ¶p013, ¶p017, ¶p018 [25] 0119 ¶p039; 0130 ¶p012 [26] 0094 ¶p030 [27] 0110 ¶p004 [28] 0083 ¶p007 [29] 0118 ¶p061 [30] 0100 ¶p063 [31] 0052 ¶p090; 0087 ¶p020; 0120 ¶p107 [32] 0083 ¶p005; 0097 ¶p008; 0113 ¶p039; 0118 ¶p063 [33] 0021 ¶p018 [34] 0055 ¶p246 [35] 0116 ¶p029, ¶p041, ¶p053 [36] 0025 ¶p233; 0033 ¶p034; 0056 ¶p026; 0075 ¶p028 [37] 0077 ¶p007 [38] 0118 ¶p062 [39] 0119 ¶p031, ¶p035 [40] 0116 ¶p017, ¶p019 [41] 0121 ¶p042

Q22: Scaling evidence for modifications; what a PCCP can cover

40 comments, 79 tags: 2 support, 70 conditional-minor, 5 conditional-major, 2 neutral

Of 79 points, 2 support, 70 ask small changes, 5 larger ones and 2 take no position. Commenters accept the premarket assessment as a baseline and evidence scaled to a change's impact [1]. Many propose tiers, usually three: interface, formatting or documentation changes stay in the quality system; bounded prompt, retrieval or guardrail changes get targeted re-benchmarking, often within a PCCP; and a new model family, intended use, population, modality, tool access or autonomy level may need premarket review [2]. Several insist the tier follow the size of the behavior change, not the code change [3]; one would tier by reversibility and blast radius [4], another by whether the locked evaluation instrument stays valid [5], and one adds cumulative small changes [6]. Several would categorize by the component changed and map each category to the benchmarking elements or competencies it can affect, so a guardrail change triggers the safety elements, a model version change everything [7]. Several would re-run a fixed safety core in full after any model or guardrail change [8], one regardless of how minor the sponsor expects the impact [9]; one wants comparison at the level of constituent failure modes, since an element pass can hide regression [10].

Commenters widen the change vocabulary: prompt, retrieval, guardrail and orchestration changes should count as device changes under change control [11]; one review of authorized PCCPs found none covering such changes [12]; one would name model compression as a category and validate the exact deployed artifact [13]; others add printer and material changes for point-of-care manufacturing [14] and verification of the sensor-to-output chain after service [15]. Several want the baseline fixed and versioned [16]: the benchmark suite never modified within the quality-system tier [17], dataset versions and partitions recorded [18], the adjudicator pinned with bridge testing when it changes [19], and evaluation variability characterised [20]; one would draw linked forms from a sequestered bank [21]. Other larger changes bind evidence to the whole deployed configuration: an immutable configuration whose every change triggers reassessment [22], an execution-time Lookback and Revalidation check that holds a function when its scope cannot be established [23], and independent revalidation of material changes before broader deployment [24]; a smaller version makes a build fingerprint the trigger [25]. One asks manufacturers to notify deployers before changes, with a recordable version identifier [26], one to clarify which hospital configurations modify a device function [27], and one to exclude open-ended foundation-model changes from PCCPs [28].

Sources. [1] 0013 ¶p130; 0052 ¶p135; 0087 ¶p031; 0127 ¶p024 [2] 0003 ¶p061; 0005 ¶p327; 0013 ¶p132; 0045 ¶p086; 0049 ¶p087; 0055 ¶p253; 0071 ¶p032; 0087 ¶p033; 0093 ¶p031; 0105 ¶p035; 0113 ¶p040; 0118 ¶p066; 0120 ¶p249; 0123 ¶p061 [3] 0025 ¶p353; 0045 ¶p085; 0052 ¶p135; 0071 ¶p031; 0120 ¶p247 [4] 0071 ¶p031 [5] 0013 ¶p130 [6] 0052 ¶p135 [7] 0067 ¶p032; 0090 ¶p035;

0091 ¶p056; 0114 ¶p033 [8] 0058 ¶p043; 0091 ¶p057; 0118 ¶p066 [9] 0058 ¶p043 [10] 0027 ¶p051 [11] 0099 ¶p021 [12] 0090 ¶p033, ¶p034 [13] 0027 ¶p005 [14] 0053 ¶p010 [15] 0032 ¶p079 [16] 0013 ¶p135; 0085 ¶p036; 0090 ¶p036; 0091 ¶p059 [17] 0013 ¶p135 [18] 0085 ¶p036; 0090 ¶p036 [19] 0091 ¶p059 [20] 0091 ¶p021 [21] 0123 ¶p063 [22] 0076 ¶p047 [23] 0105 ¶p008 [24] 0047 ¶p011 [25] 0127 ¶p036 [26] 0130 ¶p013 [27] 0035 ¶p035 [28] 0118 ¶p066

Q23: Adapting PCCPs to changes that can't be fully prespecified

21 comments, 33 tags: 16 conditional-minor, 15 conditional-major, 2 neutral

Of 33 points, 16 ask small changes, 15 larger ones and 2 take no position; none supports or opposes outright. Most commenters converge on one idea: since future changes cannot be enumerated, a PCCP should prespecify what must hold after any change rather than the change itself [1]. They call it an allowable change envelope [2], invariants with a locked, escrowed evaluation suite, thresholds and protocol [3], a fence [4], a Performance-Bounded PCCP [5], safety invariants such as evidence grounding, escalation and human override that cannot disappear without revalidation [6], predetermined gates [7], evidence-gated state transitions [8], a prespecified re-examination or standing re-benchmarking protocol [9], or procedural and boundary prespecification [10]. One notes that whether current PCCP authority allows this is a separate legal question [11]. Pushback comes from two commenters [12]: a performance envelope is a monitoring tool, not authorization for unspecified changes, since stable averages can hide subgroup deterioration [13]; and the safe default for an unspecifiable change space is fail-closed resumption, not broader prespecification [14]. Added elements: subgroup-level thresholds [15], accuracy-band tiers from no submission to full review [16], staged progression toward autonomy with downgrading on signals [17], prohibited changes and no open-ended authorization [18], environmental changes that invalidate assumptions [19], and a defined scope for memory or adaptive retrieval [20]. On learning in use, one would let corrections feed offline retraining but never let the production model learn silently [21].

Sources. [1] 0005 ¶p328; 0013 ¶p138; 0016 ¶p002; 0025 ¶p358; 0035 ¶p083; 0045 ¶p088; 0049 ¶p089; 0051 ¶p024; 0052 ¶p137; 0058 ¶p042; 0087 ¶p034; 0093 ¶p032; 0118 ¶p068; 0120 ¶p253; 0122 ¶p014 [2] 0005 ¶p328; 0051 ¶p024 [3] 0013 ¶p138, ¶p148 [4] 0045 ¶p088 [5] 0016 ¶p004 [6] 0025 ¶p359, ¶p361 [7] 0093 ¶p032 [8] 0122 ¶p014 [9] 0058 ¶p042; 0118 ¶p068 [10] 0049 ¶p089; 0052 ¶p137; 0087 ¶p034; 0120 ¶p253 [11] 0049 ¶p089 [12] 0116 ¶p045; 0131 ¶p062 [13] 0116 ¶p045, ¶p083 [14] 0131 ¶p062 [15] 0013 ¶p140; 0113 ¶p041 [16] 0113 ¶p041 [17] 0052 ¶p138 [18] 0120 ¶p254, ¶p255 [19] 0052 ¶p139 [20] 0075 ¶p035 [21] 0120 ¶p112

Q24: Changes initiated by third-party foundation model developers

44 comments, 96 tags: 91 conditional-minor, 3 conditional-major, 2 neutral

Of 96 points, 91 ask small changes, 3 larger ones and 2 take no position. Two commenters call this the paper's largest unaddressed gap or its biggest practical challenge [1], and several describe hosted models changing without notice, sometimes under stable labels, on short deprecation timelines [2]. Several hold that the manufacturer cannot disclaim a change it did not make, and that a dependency it cannot control is itself an unresolved device risk [3]. Two would treat such changes as supply-chain events or software of unknown provenance [4].

The remedies form a layered stack. Contractual terms: advance notice with a lead time long enough to re-benchmark, version disclosure, a deprecation window, access to prior versions and audit rights [5]; two name ninety or fourteen days [6]. Version pinning [7], which several want FDA to state as expected practice or a design specification, inability to pin being a design deficiency [8]; one notes pinning is not always offered, is eventually retired, and misses changes beneath a stable label [9]. Several call notice necessary but insufficient, since it describes intent, not effect, and cannot be verified [10], so many add technical detection: fixed canary sets run against the live endpoint, version fingerprints or manifests,

and drift monitoring [11]. Many want a gate before any new version reaches patients: re-benchmarking against the premarket baseline or a regression battery, sometimes in shadow mode [12]; one says a detected change carries the same evidence obligation as a chosen one [13], and one would treat a forced migration as a submission-level change [14]. Several pair this with rollback, suspension or a restricted mode when a change is detected [15].

Where pinning or notice cannot be obtained, one would raise premarket evidence [16], others narrow permitted use [17], or disclose the limit in the benefit-risk assessment [18]. Several would avoid the dependency [19]: safety-critical logic kept outside the third-party model [20], or locally hosted, hash-verified weights that leave no third-party change to detect [21]. On PCCPs, several see a place for anticipated upstream changes, such as like-for-like version upgrades that pass a prespecified regression battery [22], while the larger changes hold that a PCCP cannot cover changes the sponsor does not control and should be unavailable to devices on an unversioned endpoint, with pinning, notice and re-benchmarking made conditions of authorization instead [23]. Other points: component identification must precede change detection, since one commenter's own code called a different vendor's API than documented [24]; deployer-observed changes should count as evidence, since vendor cooperation is doubtful [25]; deployers and practices should be told of changes before or on the day they take effect [26]; data-handling changes should count as material [27]; and model, API or service retirement needs disclosed support periods and transition plans [28].

Sources. [1] 0013 ¶p149; 0018 ¶p033 [2] 0013 ¶p150; 0054 ¶p104; 0090 ¶p038, ¶p040; 0096 ¶p003 [3] 0021 ¶p019; 0025 ¶p265; 0045 ¶p089; 0055 ¶p280; 0113 ¶p042 [4] 0017 ¶p008; 0030 ¶p012 [5] 0003 ¶p064; 0013 ¶p152; 0017 ¶p041; 0021 ¶p019; 0025 ¶p267; 0035 ¶p083; 0045 ¶p090; 0049 ¶p090; 0055 ¶p264; 0093 ¶p033; 0100 ¶p066; 0102 ¶p010; 0113 ¶p042; 0116 ¶p046; 0117 ¶p009; 0118 ¶p070; 0120 ¶p259; 0123 ¶p062; 0126 ¶p213 [6] 0003 ¶p064; 0102 ¶p010 [7] 0003 ¶p065; 0006 ¶p027; 0013 ¶p151; 0018 ¶p037; 0048 ¶p009; 0049 ¶p091; 0053 ¶p010; 0055 ¶p273; 0076 ¶p044; 0100 ¶p066; 0105 ¶p036; 0116 ¶p046; 0120 ¶p259; 0123 ¶p062; 0126 ¶p213 [8] 0006 ¶p027; 0013 ¶p151; 0100 ¶p066; 0126 ¶p213 [9] 0090 ¶p040 [10] 0062 ¶p050; 0090 ¶p039; 0131 ¶p061 [11] 0005 ¶p329; 0013 ¶p153; 0035 ¶p083; 0048 ¶p009; 0049 ¶p090; 0062 ¶p040; 0090 ¶p041; 0093 ¶p033; 0096 ¶p011; 0105 ¶p036; 0118 ¶p070; 0120 ¶p259; 0127 ¶p036; 0131 ¶p061 [12] 0003 ¶p065; 0006 ¶p027; 0052 ¶p084; 0055 ¶p279; 0076 ¶p044; 0086 ¶p010; 0090 ¶p043; 0100 ¶p067; 0118 ¶p070; 0123 ¶p058; 0126 ¶p213 [13] 0090 ¶p043 [14] 0013 ¶p154 [15] 0025 ¶p267; 0049 ¶p091; 0055 ¶p278; 0075 ¶p040; 0086 ¶p010; 0105 ¶p036; 0120 ¶p262; 0130 ¶p013 [16] 0003 ¶p065 [17] 0048 ¶p009; 0075 ¶p038 [18] 0120 ¶p261 [19] 0082 ¶p034; 0096 ¶p007; 0100 ¶p065 [20] 0100 ¶p065 [21] 0082 ¶p034; 0096 ¶p007 [22] 0006 ¶p027; 0100 ¶p068; 0120 ¶p260; 0126 ¶p214 [23] 0062 ¶p042; 0076 ¶p044 [24] 0062 ¶p008, ¶p017 [25] 0131 ¶p058 [26] 0110 ¶p005; 0118 ¶p070; 0130 ¶p013 [27] 0101 ¶p010 [28] 0032 ¶p067; 0075 ¶p040

MASTER_FILES: Foundation model master files

37 comments, 98 tags: 10 support, 52 conditional-minor, 28 conditional-major, 8 neutral

Q25: Voluntary Foundation Model Master Files

37 comments, 98 tags: 10 support, 52 conditional-minor, 28 conditional-major, 8 neutral

Of 98 points, 10 support, 52 ask small changes, 28 larger ones and 8 take no position. Supporters see the files sharing upstream information and cutting duplicative review [1]. Many hold, with the paper, that a file is no authorization and never substitutes for validating the configured device, for which the sponsor stays responsible [2].

The split is over the incentive problem. The larger changes would give the program teeth: mandatory disclosure for models in regulated or higher-consequence devices [3]; public model cards, since a model's limitations are labeling, not trade secrets [4]; or a default under which, absent an adequate file, the device maker carries the full characterization or a higher premarket evidence burden [5]. One would expect a file wherever a sponsor relies materially on a third-party model [6]; another would pair the file with contractual disclosure duties and FDA authority to ask for more [7]. The smaller changes work inside the voluntary design: more predictable review for devices on filed models [8], letting a referenced file satisfy part of a sponsor's benchmarking or re-examination burden [9], procurement pull [10], and a sponsor-side floor: the sponsor documents the deployed model's identity, version and change behavior itself [11]; one insists a missing file create no presumption against a device [12], another that FDA require another method where a provider withholds [13]. Alternatives named: supplier contracts and purchasing controls, suppliers producing sensitive material directly to FDA without discharging the manufacturer's duty [14], platform intermediaries as file holders [15], and structured system cards, version APIs, attestations and third-party audits [16].

Content lists converge: model identity and version lineage, training-data provenance, supported and unsupported uses, limitations and failure modes, subgroup results, guardrails and refusal behavior, update-notification and deprecation commitments, and audit-log availability [17]. Several say a file is useful only if current and version-specific [18]; two would make versioning, deprecation and change-notification practices core content [19]. Additions: the filer's own detection methodology, since a card discloses only what the filer knows [20]; content-policy changes between versions and output stability under user framing [21]; stability under quantization and other downstream changes, since the file describes the model as produced, not deployed [22]; a write-authority declaration [23]; and tiered disclosure protecting proprietary detail [24], or conversely a narrow public tier of versions, change dates and safety events [25].

Two commenters question the document itself. One measured a hosted assistant's model exchanged 153 times in fourteen hours and concludes no filing can stay current: limit the file to stable facts, require execution locality and a machine-readable version the serving system reports, and treat the identity of the answering model as a continuous record, user-side or device-held [26]. Another would build the file on hash commitments to specific evaluation results rather than documents, and scale expectations by whether a model can be independently re-tested [27].

Sources. [1] 0012 ¶p010; 0054 ¶p095; 0087 ¶p036; 0100 ¶p070; 0120 ¶p122; 0126 ¶p217 [2] 0013 ¶p164; 0025 ¶p366; 0046 ¶p020; 0049 ¶p095; 0054 ¶p096; 0055 ¶p132; 0087 ¶p038; 0099 ¶p017; 0105 ¶p038; 0114 ¶p026; 0119 ¶p055; 0120 ¶p268; 0126 ¶p217 [3] 0106 ¶p018; 0113 ¶p044 [4] 0076 ¶p041 [5] 0003 ¶p069; 0013 ¶p160; 0076 ¶p040 [6] 0017 ¶p043 [7] 0045 ¶p092 [8] 0006 ¶p027 [9] 0058 ¶p047; 0118 ¶p072 [10] 0093 ¶p034 [11] 0005 ¶p330; 0093 ¶p034; 0100 ¶p071 [12] 0100 ¶p071 [13] 0055

¶p133 [14] 0115 ¶p043; 0119 ¶p052, ¶p053 [15] 0100 ¶p072 [16] 0049 ¶p096; 0120 ¶p269 [17] 0003 ¶p070; 0005 ¶p330; 0013
¶p163; 0017 ¶p043; 0035 ¶p085; 0049 ¶p094; 0054 ¶p096; 0055 ¶p120; 0087 ¶p037; 0105 ¶p037; 0115 ¶p042; 0120 ¶p265 [18] 0005
¶p330; 0013 ¶p163; 0049 ¶p093; 0105 ¶p037 [19] 0013 ¶p155; 0032 ¶p102 [20] 0008 ¶p011 [21] 0058 ¶p045 [22] 0027 ¶p071
[23] 0127 ¶p043 [24] 0035 ¶p085 [25] 0050 ¶p016 [26] 0079 ¶p005, ¶p008, ¶p032, ¶p035, ¶p038, ¶p064 [27] 0131 ¶p065, ¶p068

AGENTIC: Agentic AI systems

52 comments, 175 tags: 1 support, 129 conditional-minor, 31 conditional-major, 14 neutral

Q26: Agentic AI systems

52 comments, 175 tags: 1 support, 129 conditional-minor, 31 conditional-major, 14 neutral

Of 175 points, 1 supports, 129 ask small changes, 31 larger ones and 14 take no position. The one support endorses the agentic benchmarking element's competencies as drafted [1]. Many hold that an agent must be judged on its whole action chain, not its final output, because errors compound across steps [2]; one wants acceptance criteria stated for the completed task [3], another per-action floors, since instability varied eightfold across actions [4], a third the end state of connected systems checked [5].

The commonest ask is a human checkpoint before any irreversible or high-consequence action [6]. Several add that it must be non-bypassable in the deployed configuration, not satisfiable by another model or by conversational wording [7], that the reviewer must be domain-matched, independent and given enough information and time [8], and that autonomous execution is allowed only where specifically evaluated [9]. The larger changes harden this: oversight redefined as pre-action authorization [10]; checkpoints required, not optional, for physical outputs [11]; oversight accepted only as a verifiable, tamper-evident record that a present human authorized each consequential action, proof at the action rather than identity at login [12]; gates acting on witnessed state, stops as reflexes outside the agent's deliberation and a log the agent cannot write [13].

Authority is the second theme: capability does not confer authority, so planning or recommending must stay distinct from permission to execute [14]. Several want a declared authority envelope or inventory of permitted actions and tools [15]; many want least-privilege, allowlisted tools [16], enforced in surrounding software rather than prompts [17]. Larger changes propose graduated authority: a five-level clinical agency ladder [18], governed reliance separated from governed execution with non-binary dispositions [19], and permission tiers declared at design time with the highest default-denied and capability expansion a human-authorized transition [20]; smaller ones scale criteria and checkpoint density to autonomy [21]. Others add action budgets or limits on cumulative actions [22] and tests for scope expanding mid-session [23].

Testing and records: prompt injection through retrieved content and tool outputs [24]; tool failure, partial execution and duplicate actions [25]; delegation between agents [26]. One finds the agentic element blind to the truthfulness of step reports, self-written user state, claims of absence and cross-session drift, and offers criteria such as scoring reports against an independent execution record and re-evaluating with the instruction layer removed [27]. Several want audit logs able to reconstruct the trajectory [28], one as an explicit premarket element [29], with attribution emitted rather than inferred [30], skipped checkpoints never reading as passed, and each action committing to its input origins and sequence length [31]. Several want a stop, kill switch or fail-safe state [32]; two make a kill switch a required larger change, one drawing on cell and gene therapy's deactivation requirements [33]. Structural proposals: a registration spine making the agent, its models, tools and datasets a reviewable configuration [34]; end-to-end responsibility where components come from different organizations [35]; judging an agent that controls another device against that device's risk [36]; presumptive exclusion from the reduced-premarket-evidence pathway [37]. Several urge that agentic orchestration alone not make a non-device function a device [38]. One notes that for nonverbal users the opportunity for review is absent by definition [39].

Sources. [1] 0061 ¶p042 [2] 0005 ¶p331; 0013 ¶p166; 0018 ¶p030; 0025 ¶p276; 0027 ¶p079; 0035 ¶p087; 0048 ¶p010; 0052 ¶p145; 0056 ¶p027; 0071 ¶p028; 0075 ¶p011; 0087 ¶p047; 0095 ¶p046; 0105 ¶p039; 0115 ¶p044; 0118 ¶p075; 0120 ¶p272 [3] 0013 ¶p166 [4] 0095 ¶p046 [5] 0075 ¶p011 [6] 0003 ¶p073; 0013 ¶p168; 0017 ¶p045; 0018 ¶p031; 0035 ¶p089; 0046 ¶p020; 0048 ¶p010; 0049 ¶p099; 0053 ¶p012; 0054 ¶p099; 0055 ¶p147; 0056 ¶p028; 0067 ¶p034; 0076 ¶p063; 0087 ¶p046; 0089 ¶p008; 0097 ¶p006; 0098 ¶p045; 0100 ¶p074; 0102 ¶p010; 0105 ¶p040; 0113 ¶p046; 0115 ¶p046; 0117 ¶p010; 0118 ¶p075; 0120 ¶p273 [7] 0056 ¶p028; 0067 ¶p033; 0100 ¶p074; 0102 ¶p010; 0117 ¶p010 [8] 0048 ¶p010; 0075 ¶p013; 0094 ¶p068; 0098 ¶p045 [9] 0054 ¶p099; 0087 ¶p046; 0105 ¶p040; 0120 ¶p273 [10] 0003 ¶p073 [11] 0053 ¶p012 [12] 0089 ¶p002, ¶p004, ¶p018 [13] 0127 ¶p042 [14] 0029 ¶p018; 0049 ¶p009; 0087 ¶p045; 0092 ¶p019; 0122 ¶p025 [15] 0013 ¶p168; 0035 ¶p088; 0049 ¶p099; 0056 ¶p028; 0115 ¶p045; 0130 ¶p014 [16] 0013 ¶p169; 0035 ¶p089; 0048 ¶p010; 0055 ¶p144; 0093 ¶p035; 0105 ¶p040; 0115 ¶p046 [17] 0061 ¶p055; 0075 ¶p012; 0093 ¶p035 [18] 0045 ¶p094 [19] 0049 ¶p102 [20] 0122 ¶p024, ¶p026 [21] 0051 ¶p025; 0052 ¶p148; 0054 ¶p097; 0089 ¶p017; 0093 ¶p035 [22] 0003 ¶p074; 0075 ¶p014; 0105 ¶p040 [23] 0071 ¶p034 [24] 0027 ¶p076; 0035 ¶p089; 0056 ¶p029; 0075 ¶p015; 0100 ¶p075; 0115 ¶p046 [25] 0052 ¶p145; 0056 ¶p029; 0075 ¶p016; 0100 ¶p076 [26] 0017 ¶p045; 0035 ¶p089; 0075 ¶p014 [27] 0061 ¶p047, ¶p049, ¶p055 [28] 0013 ¶p171; 0017 ¶p045; 0054 ¶p099; 0067 ¶p034; 0075 ¶p048; 0076 ¶p063; 0093 ¶p035; 0115 ¶p045; 0130 ¶p014 [29] 0013 ¶p171 [30] 0042 ¶p047 [31] 0131 ¶p072, ¶p074, ¶p075 [32] 0003 ¶p075; 0017 ¶p045; 0022 ¶p007; 0035 ¶p089; 0055 ¶p150; 0100 ¶p076; 0105 ¶p040; 0113 ¶p046; 0115 ¶p046; 0130 ¶p014 [33] 0109 ¶p013; 0118 ¶p076 [34] 0122 ¶p007, ¶p009 [35] 0052 ¶p144 [36] 0051 ¶p025 [37] 0013 ¶p172 [38] 0094 ¶p070; 0100 ¶p073; 0120 ¶p275 [39] 0061 ¶p011